

Responsive neuromodulation in temporal lobe epilepsy: Preclinical investigation of the safety and efficacy of the WiChR opsin as an optogenetic tool

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Preface

This master's dissertation marks the end of our academic journey. It has been both a challenging and rewarding, and we are deeply grateful to all those who have supported us along the way.

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1 Summary

Background: Temporal Lobe Epilepsy (TLE) is the most common and drug-resistant form of epilepsy, emphasizing the need for alternative treatments. Responsive neuromodulation (RNM) using optogenetics is a promising approach that employs genetically encoded, light-sensitive opsins to modulate neural activity in real-time with high temporal and spatial precision. WiChR, a novel inhibitory opsin, holds potential for seizure suppression, but its safety and efficacy remain to be validated.

Methods: A dose-titration study was conducted using three dilutions (1/5, 1/10, 1/100) of the AAV vector encoding WiChR to assess neurotoxicity in the mouse hippocampus. Histological evaluation focused on markers of hippocampal degeneration and microglial activation. The safest dilution (1/100) was subsequently used in the intrahippocampal kainic acid (IHKA) mouse model for TLE. This dilution was then used in a RNM experiment to assess seizure suppression.

Results: A dose-dependent increase in hippocampal degeneration and microgliosis was observed at higher viral titers (1/5, 1/10), while the 1/100 dilution showed no signs of toxicity. The addition of WiChR to the IHKA mouse model did not induce neurotoxicity beyond that caused by KA alone. Following improved WiChR expression using the Nanoliter injector instead of Hamilton, optogenetic activation significantly reduced seizure duration.

Conclusion: This study identified a safe and effective AAV-WiChR dose that enables seizure suppression in a mouse model of TLE, thereby validating the potential of WiChR-based neuromodulation. These findings support further investigation into WiChR for clinical application and highlight the importance of dosing precision and delivery technique in optogenetic therapy.

2 Societal impact

Epilepsy affects millions of people worldwide, and many patients continue to experience uncontrolled seizures despite treatment. While anti-seizure medication (ASM) is the standard therapeutic approach, it is frequently associated with significant adverse effects, including fatigue, memory impairments, mood disturbances, and a diminished quality of life.

Our research offers a novel, more targeted alternative: responsive neuromodulation through optogenetics. This technique involves a brain implant capable of detecting and suppressing seizure activity in real-time using light-sensitive proteins. This approach introduces two key advantages over current therapeutic strategies. First, unlike electrical neurostimulation, which affects broader brain regions, optical stimulation selectively modulates the seizure-generating neurons. Second, in contrast to continuous stimulation, responsive treatment exclusively delivers intervention during seizure events. The combination of this high spatial and temporal precision and accuracy holds the potential to reduce the incidence of side effects and thereby improve quality of life.

This technology holds particular promise for individuals with drug-resistant epilepsy, especially those ineligible for surgical intervention or who experience debilitating side effects from ASM. The envisioned impact extends beyond healthcare and biotech sectors, offering long-term benefits for patients, neurologists, and the broader medical community. Stakeholders include patients and their families, physicians, healthcare systems, and patient advocacy groups.

This research aims to investigate optogenetics as a novel treatment method, ultimately reducing the burden of epilepsy on individuals and healthcare systems, improving long-term outcomes, enhancing social inclusion and quality of life for people with chronic neurological conditions, and paving the way for future precision neurotherapies.

3 Introduction

3.1 Epilepsy

3.1.1 Definition & Classification

Epilepsy is one of the most common neurological disorders worldwide, affecting around 50 million people or 1% of the global population¹. The disease is characterized by the recurrence of unprovoked seizures, defined as periods of abnormal, excessive and hypersynchronous neural activity. It involves sudden changes in the level of motoric function, consciousness, behavior, memory or emotions².

Epilepsy can be classified according to seizure onset, origin, and symptoms. Currently, there are three distinct categories of seizure onset – *generalized*, *focal*, or *unknown*. Each is based on how and where the seizure begins in the brain. *Primary generalized onset* involves the rapid and widespread activation of both cortical hemispheres. *Focal* or *partial onset* seizures start off in one area; however, it may rapidly spread and create a *secondary generalized seizure*. *Focal* seizures can start off in any lobe of the brain, of which the mesial temporal lobe, containing the hippocampus and amygdala, is most common. This form is known as temporal lobe epilepsy (TLE). Lastly, epilepsy may be categorized based on symptoms, such as absence seizures, tonic-clonic seizures, and status epilepticus¹.

3.1.2 Etiology and Pathophysiology

As the definition of epilepsy suggests, the patient must experience at least two unprovoked seizures, occurring more than 24 hours apart¹. The cause of epilepsy can range from complications during pregnancy and childbirth to brain tumors. Generally, the cause can be attributed to an alteration in the brain circuitry, resulting in excitation/inhibition imbalance. Neurons maintain this balance by communicating through a combination of electrical and chemical signals. In healthy circumstances, an action potential (AP) is created because stimuli, such as neurotransmitters, allow ion channels to open and sodium ions to enter the cell. This influx of sodium ions increases the membrane potential, eventually triggering voltage-dependent sodium channels at the threshold of -55 mV. The rapid influx of positive ions, called depolarization, generates the AP. After the peak, the neuron enters a repolarization phase, where sodium channels close and potassium channels open, allowing an efflux of positive potassium ions. The Na⁺/K⁺-ATPase restores the equilibrium of sodium and potassium ions, preparing for the next AP (Figure 1). This process occurs sequentially along the neuron's membrane – primarily the axon – where each AP stimulates the start of a new AP further down the axon. At the synapse, depolarization will cause the release of neurotransmitters. These chemical messengers can either activate or inhibit the post-synaptic neuron, depending on the neurotransmitter and receptor involved. During a seizure, the brain experiences a “short circuit” where connections become overactive, causing neurons to exert excessive discharges that propagate throughout the brain. Symptoms and signs may vary, depending on which regions are affected.

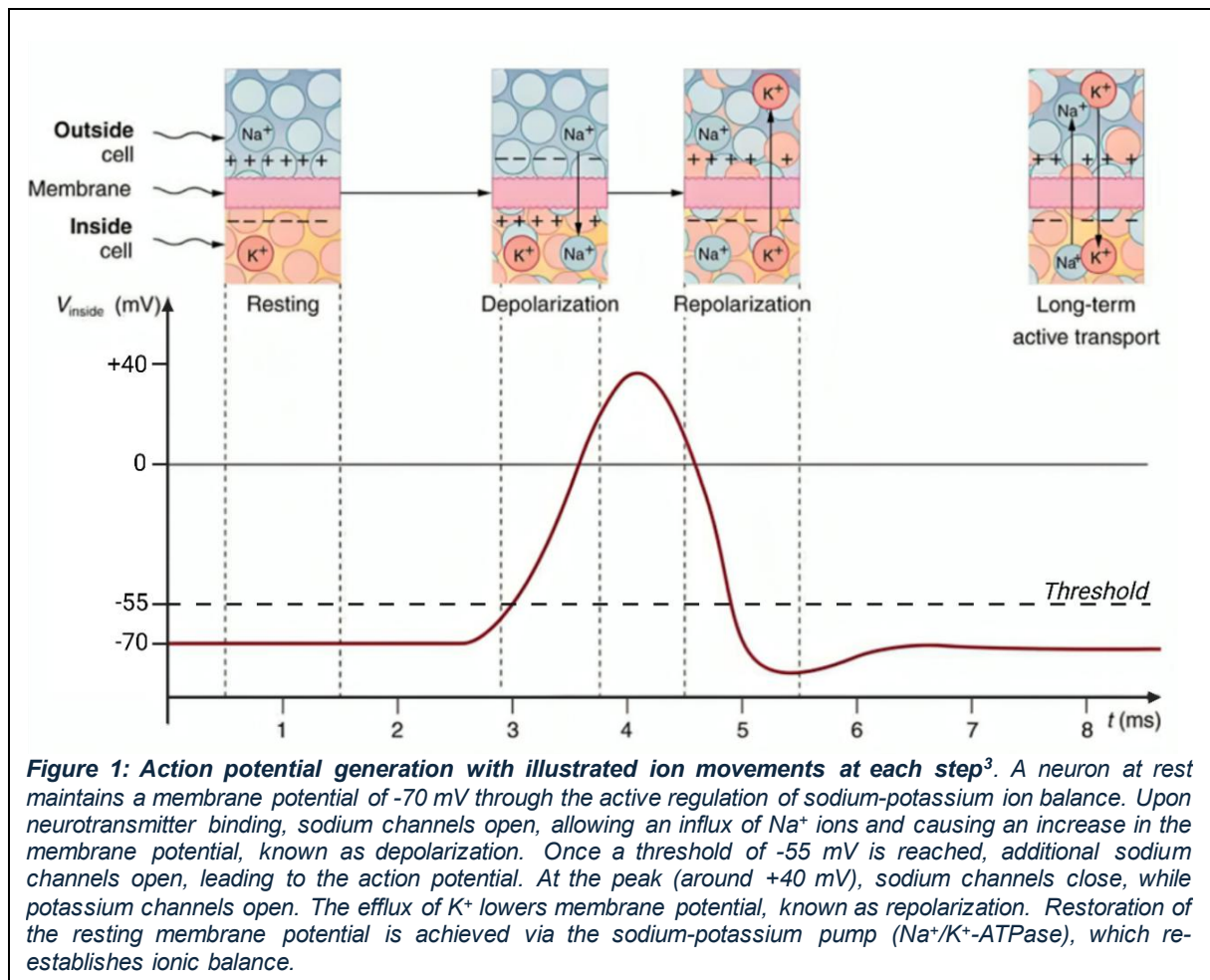


Figure 1: Action potential generation with illustrated ion movements at each step³. A neuron at rest maintains a membrane potential of -70 mV through the active regulation of sodium-potassium ion balance. Upon neurotransmitter binding, sodium channels open, allowing an influx of Na^+ ions and causing an increase in the membrane potential, known as depolarization. Once a threshold of -55 mV is reached, additional sodium channels open, leading to the action potential. At the peak (around +40 mV), sodium channels close, while potassium channels open. The efflux of K^+ lowers membrane potential, known as repolarization. Restoration of the resting membrane potential is achieved via the sodium-potassium pump (Na^+/K^+ -ATPase), which re-establishes ionic balance.

As mentioned, age also plays an important role in the etiology of epilepsy, with the highest incidence observed in both the youngest and oldest age groups². In the developing brain, increased susceptibility to seizures is partly due to the excitatory action of the inhibitory neurotransmitter gamma-aminobutyric acid (GABA) during early development; GABAergic synapses form prior to glutaminergic ones^{3,4}. In older individuals, the elevated risk of epilepsy is largely attributable to the high prevalence of cerebrovascular disease and associated neuronal damage². In addition to acquired causes, genetic factors also play a crucial role in the development of epilepsy. Mutations affecting receptors and ion channels, as well as genetic variants underlying neurodevelopmental disorders such as cortical dysplasia, are frequently implicated in epilepsy. The understanding of the genetic architecture of epilepsy continues to advance, with both inherited and *de novo* variants – within coding and non-coding regions – being recognized as contributors to seizure disorders^{5, 6, 7} (Table 1, Common Causes of Epilepsy¹).

Age Group	Common Causes
Infants	Perinatal hypoxia, metabolic disorders, intracranial hemorrhage, genetic disorders, developmental and congenital maldevelopment
Children	Perinatal anoxia, injury at birth or later, infection, vascular, metabolic, cortical malformations, genetic disorders
Teens & young adults	Trauma, infection, genetic disorders, brain tumor, congenital
Adults	Stroke
Older adults (>60 years)	Stroke, brain tumor, infection, trauma

Table 1: Common causes of epilepsy¹.

3.1.3 Treatments

First-line treatment for epilepsy consists of anti-seizure medication (ASM). ASM can be categorized based on its mechanism of action and the types of seizures it is effective against. In general, they can be divided into narrow spectrum ASM and broad-spectrum. As the names suggest, narrow-spectrum ASM is effective only against focal seizures, and broad-spectrum ASM is effective against a wide variety of seizures. Treatment generally starts with broad-spectrum ASM as a safety measure when diagnosis is uncertain. The mechanism of action varies among ASM, but typically it acts as a blocker for excitatory voltage-gated sodium and calcium channels (valproate, lamotrigine, oxcarbazepine, cenobamate...), enhances hyperpolarizing potassium channel function (lacosamide, rufinamide), and/or enhances the inhibitory effect of GABA (valproate, gabapentin, vigabatrin, cenobamate...) ^{4, 8}. Some ASM may act as an antagonist of the post-synaptic α -amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (AMPA) receptor or N-methyl-D-aspartate (NMDA) receptor, such as perampanel and topiramate, respectively ^{8, 9}. There is even new ASM that acts as a 5-hydroxytryptamine (serotonin) receptor agonist – fenfluramine ⁹.

Although ASM is primarily developed to control seizure activity, it exerts broader effects on both the brain and the body (Table 2). Common adverse effects associated with most ASM include fatigue, headache, dizziness, and nausea. Certain drugs, such as valproate and felbamate, are also known to cause loss of appetite, which can lead to anorexia. Behavioral side effects, including mood alterations, irritability, and depression, have been reported with drugs like perampanel and levetiracetam. Furthermore, memory disturbances – already linked to epilepsy itself – can manifest during treatment with agents like topiramate, oxcarbazepine, and phenytoin. Importantly, side effects are not confined to the Central Nervous System (CNS) but may affect other organs as well. Serious toxicities, including liver toxicity, anemia, high blood pressure, cardiovascular complications, and organ failure, have been documented. Moreover, ASM is often discontinued in pregnant women due to concerns regarding its potential teratogenic effects on the fetus ⁸. It should be noted, however, that side effects vary considerably among individuals and can often be mitigated through careful dosage control and monitoring.

Side effects	Anti-seizure medication
Fatigue, headache, dizziness, & nausea	Felbamate – rufinamide – perampanel – oxcarbazepine – eslicarbazepine acetate – cenobamate
Decreased appetite and weight-loss	Felbamate – phenytoin – valproate – stiripentol
Behavioral side effects Alternate mood states Irritability Depression	Felbamate – vigabatrin – gabapentine – zonisamide – perampanel – levetiracetam – brivacetam
Memory disorders and cognitive impairments	Topiramate – phenytoin – oxcarbazepine – tiagabine
Systemic effects Liver toxicity Anemia Cardio-vascular effects Organ failure	Felbamate – phenytoin – vigabatrin – gabapentin – lacosamide
Teratogenic effects and embryo death	Valproate, brivaracetam

Table 2: Overview of common side effects associated with ASM. A simplified summary of various side effects and the corresponding ASM, adapted from review by Akyüz et al. ⁸

Although ASM is effective for a majority of patients, approximately 30% of individuals with epilepsy do not respond adequately to pharmacological treatment – a condition referred to as refractory or drug-resistant epilepsy. Focal epilepsies are most prone to being refractory, with TLE being the most common subtype¹⁰. This clinical challenge necessitates the exploration and implementation of alternative therapeutic strategies. Surgical intervention remains the most effective treatment for refractory epilepsy, with over 60% of patients achieving seizure-freedom following surgery¹¹. Therefore, drug-resistant patients should be evaluated for possible surgical treatment through comprehensive presurgical investigations that localize the epileptogenic zone^{1, 12}. Surgical approaches include open resection, MRI-guided laser ablation and, various forms of (hemi-)disconnection^{1, 8}. However, not all patients are eligible for surgery, particularly when the epileptogenic focus cannot be accurately localized or when resection poses a high risk of functional impairment¹³. Another treatment option for patients with refractory epilepsy is dietary therapy, which is often used in children. The ketogenic diet – a high-fat, low-carbohydrate regimen – has been proven to be an effective method of treatment^{1, 8, 14}.

Neuromodulation represents an additional treatment strategy for drug-resistant epilepsy, involving the modulation of nerve activity through targeted stimulation, including Vagus Nerve Stimulation (VNS) and Deep Brain Stimulation (DBS). VNS involves the surgical placement of an electrode around the vagus nerve in the neck region. The mechanism of action is hypothesized to involve an increase in inhibitory neurotransmission and/or desynchronization of brain networks, thereby suppressing seizures¹⁵. DBS requires the implantation of a depth electrode in a specific brain region. It was originally developed for the treatment for Parkinson's Disease (PD), where the electrodes typically target structures within the basal ganglia. In epilepsy, the choice of stimulation target depends on the type of epilepsy and the location of the epileptogenic focus. The mechanism of action of DBS was thought to be inhibitory in nature, similar to that of an ablative procedure. However, it is now recognized as more complex than purely inhibitory¹⁶.

Both VNS and DBS are established and regulatory-approved therapies for drug-resistant epilepsy, with VNS approved since the 1990s (Europe: 1994; USA: 1997) while DBS targeting the anterior nucleus of the thalamus (ANT) received approval in the 2000s (Europe: 2010; USA: 2018)^{17, 18, 19}. Despite their approval, the efficacy of VNS remains limited, only working in less than 10% of drug-resistant epileptic patients. Although ANT-DBS showed promise for treating epilepsy, its continuous stimulation of broad brain regions can increase the risk of adverse effects. To address these limitations responsive neurostimulation (RNS) was introduced, only available in the USA (2013)¹⁶. RNS differs from DBS by continuously monitoring intracranial electroencephalogram (EEG) activity and delivering brief, localized electrical stimulation directly to the epileptogenic zone via precisely implanted electrodes. However, electrical stimulation still affects larger brain structures.

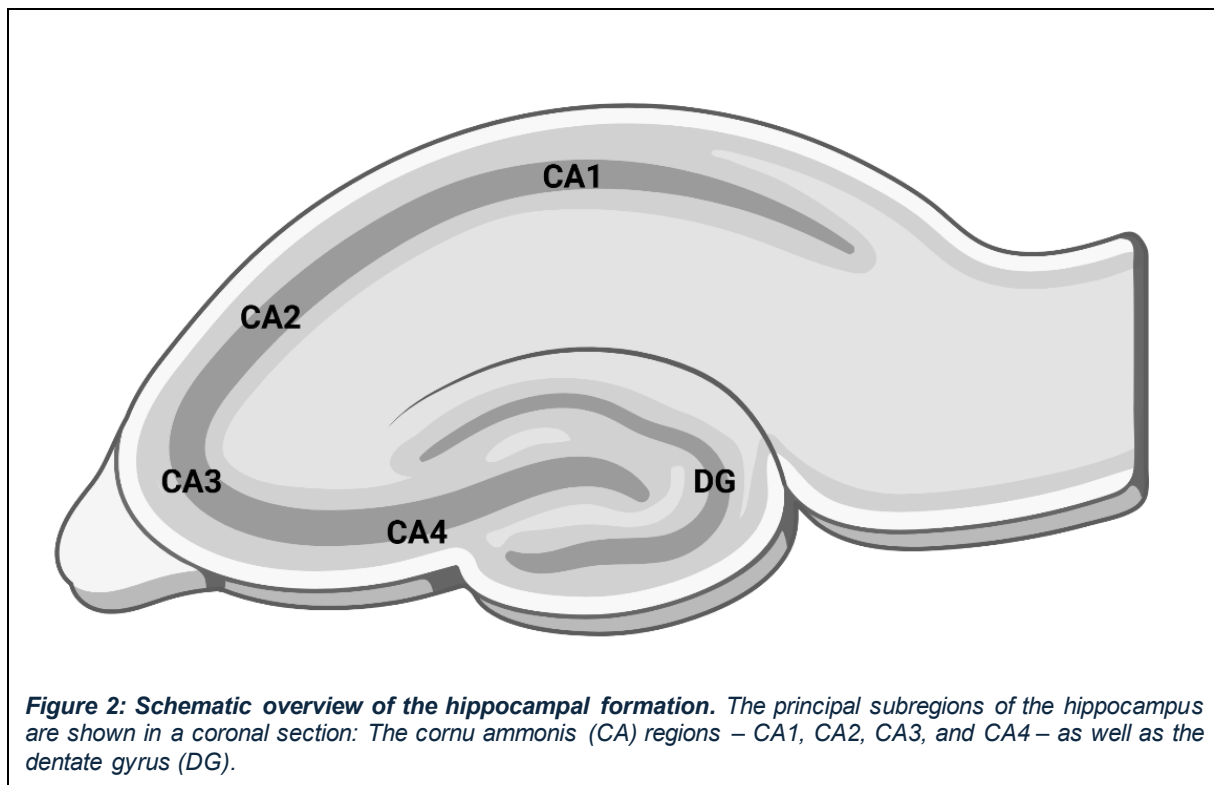
The technique of interest in this study, responsive neuromodulation using optogenetics, works on the cellular level by implementing a combination of two highly precise techniques, gene therapy and optical stimulation. In epilepsy, optogenetics suppresses seizures by activating virally introduced light-sensitive opsins through a light-emitting probe. These opsins are only expressed on specific neurons, making optogenetics extremely accurate. Stimulating these opsins through controlled light pulses is less detrimental than electric pulses, decreasing risk of damage and adverse effects. This technique creates a very accurate, adaptable therapy against epilepsy, while largely preserving brain structure and function²⁰. This study employs a responsive design, enhancing temporal accuracy. Whereas continuous design delivers constant light stimulation, persistently inhibiting (hyperexcited) neurons, the responsive design activates the light probe only in response to a seizure onset²¹. This technique shows great promise since it only targets the transfected neurons in the epileptogenic zone, as opposed to the whole brain when using ASM and thus limiting the side-effects. Further studies have shown that optogenetic silencing of epileptiform activity is successful in chronic models of focal epilepsy, including animal models of Temporal Lobe Epilepsy (TLE)²².

3.2 Intrahippocampal Kainic Acid Mouse Model

The development of treatment modalities starts with a preclinical phase where the treatment modality is tested on animals. A critical factor in this phase is the existence of a model that resembles a certain human disorder/disease as closely as possible. In Temporal Lobe Epilepsy (TLE) patients, key electrophysiological abnormalities include Spontaneous Recurrent Seizures (SRSs), interictal spikes, and high-frequency oscillations²³. These are accompanied by prominent pathological hallmarks, such as hippocampal sclerosis, granule cell dispersion, and mossy fiber sprouting¹⁰. The aim of a preclinical model is to develop a chronic condition in animals that replicates these defining characteristics, namely the SRSs, which constitute the core characteristic of epilepsy²³.

Chemoconvulsants are commonly used to induce epileptogenesis and can be administered systemically or locally into the brain²³. Among these, pilocarpine and Kainic Acid (KA) are the most commonly employed. Pilocarpine is a muscarinic acetylcholine receptor agonist and was first utilized to induce experimental epilepsy in rats. A key distinction between the pilocarpine and KA model lies in the speed of neuronal damage. After administration of pilocarpine, neuronal damage is detectable within three hours of Status Epilepticus (SE) onset, while this takes eight hours in case of KA. Another widely utilized approach for modelling TLE involves various forms of electrical stimulation. The most prominent among these is the electrical kindling model, which entails implanting an electrode in the brain to establish a seizure focus by repeated electrical stimulation. This study utilizes a KA model, because the pilocarpine & kindling models have several limitations: the pilocarpine model frequently leads to extensive extrahippocampal damage, and the kindling model does not consistently induce chronic spontaneous seizures or hippocampal sclerosis. Therefore, the KA model is best suited to reliably reproduce the key features of TLE.

KA is a cyclic analogue of L-glutamate and an agonist of the ionotropic kainite receptors (KARs)¹⁰. It was first reported to induce damage to hippocampal pyramidal neurons²³. Later research showed that increased doses of KA led to focal seizures, which subsequently evolved into SE, demonstrating the potential to use KA as a preclinical model for epilepsy. The administration of KA results in the activation of its receptors, contributing to the observed neurotoxic and epileptogenic effects. KARs are expressed throughout the brain in several regions with a particularly abundant expression in the hippocampus. KARs belong to the ionotropic glutamate receptor family, and they are involved in excitatory neurotransmission. Five subunits of KARs have been identified: GluR5 (GluK1), GluR6 (GluK2), GluR7 (GluK3), KA1 (GluK4), and KA2 (GluK5). GluK4 is predominantly localized in the CA3 region of the hippocampus, while GluK5 is expressed in both CA1 and CA3, cortex, and striatum (Figure 2, hippocampal regions)¹⁰. The high affinity of these subunits for KA and their dense concentration in CA3 may account for the damage observed in this area, suggesting a pivotal role for these subunits in mediating KA-induced neuronal cell death. Even though some subunits have higher affinity for KA than others, KA administration interacts with all types of KA receptors, exerting a particularly pronounced effect within the hippocampus²³. The mechanism behind KA-induced epilepsy is excitotoxicity, a pathological process in which neurons suffer severe damage, ultimately leading to cell death, as a result of excessive stimulation by excitatory neurotransmitters such as glutamate.



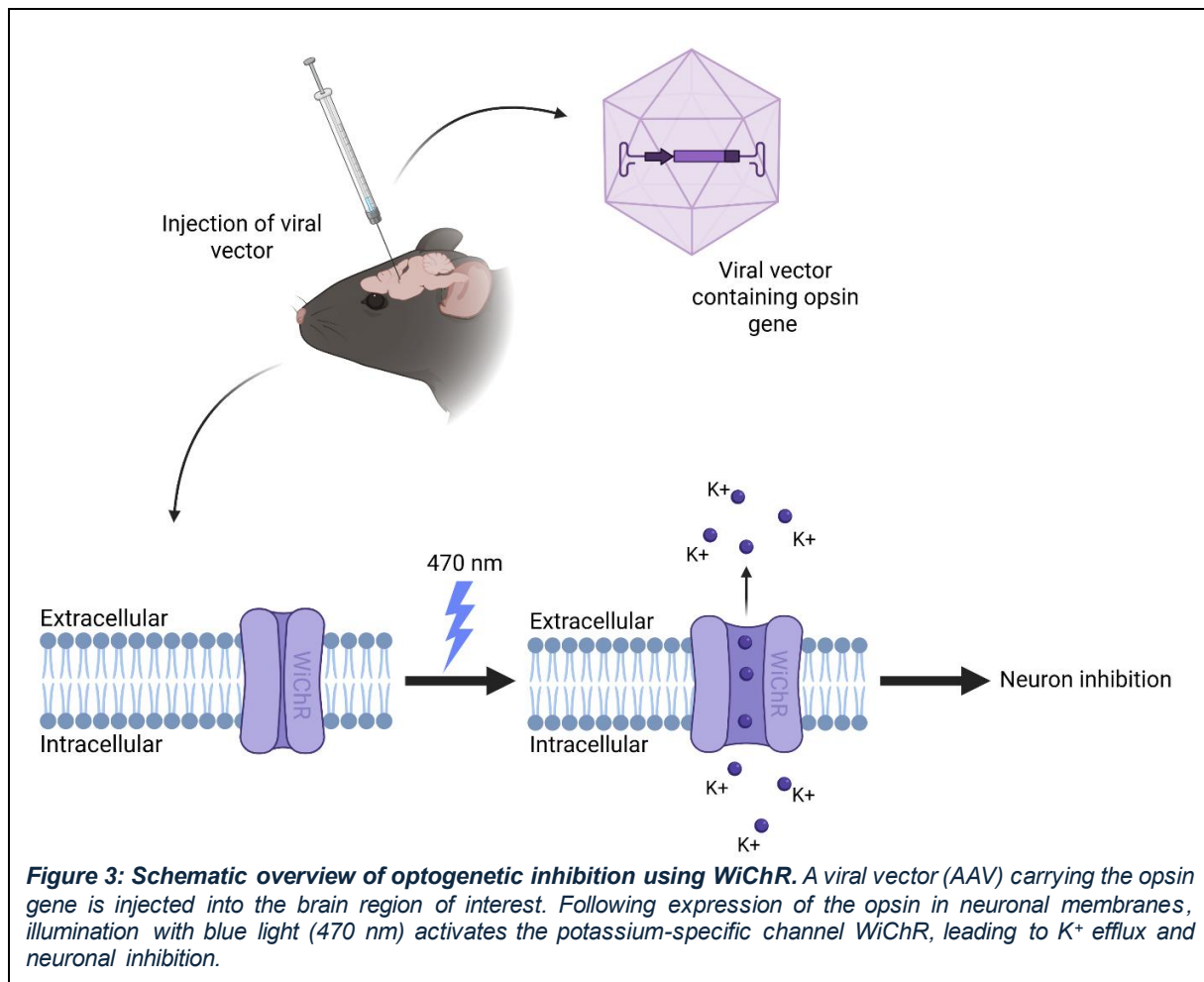
KA can be administered in various ways, including systemic, intracerebral (intraventricular, intrahippocampal, suprahippocampal, intra-amygdaloid), and intranasal²³. Intrahippocampal KA (IHKA) injection has two main advantages: (1) the blood-brain-barrier is directly bypassed and (2) the injection site is well known and controlled, leading to an increased reproducibility in lesion formation and location. Additionally, the introduced damage is local, leading to a reduction in mortality rates compared to systemic administration. Furthermore, the IHKA model is well known within literature, facilitating reproducibility across research institutions. Mice typically undergo SE within 30 minutes after KA injection.

Histological evaluation shows that IHKA injection induces a focal lesion within the hippocampus, predominantly affecting the ipsilateral CA3 subfield, but also CA1 and CA4. The KA lesion also induces granule cell dispersion in the dentate gyrus, which means that the granule cells become dispersed into adjacent areas instead of being tightly packed in the granule cell layer. Additionally, it causes mossy fiber sprouting, a phenomenon often observed in patients with temporal lobe epilepsy²³.

3.3 Optogenetics

3.3.1 History

Optogenetics is a technique that combines optical and genetic approaches to enable the precise activation or inhibition of specific cellular events within defined cell populations in living tissues²⁴. Light is employed to modulate neuronal circuits by stimulating or inhibiting them through the delivery of gene constructs encoding of protein channels (Figure 3)²⁵. This methodology was developed to address a critical challenge in neuroscience: the ability to selectively manipulate the activity of a single cell type²⁴.



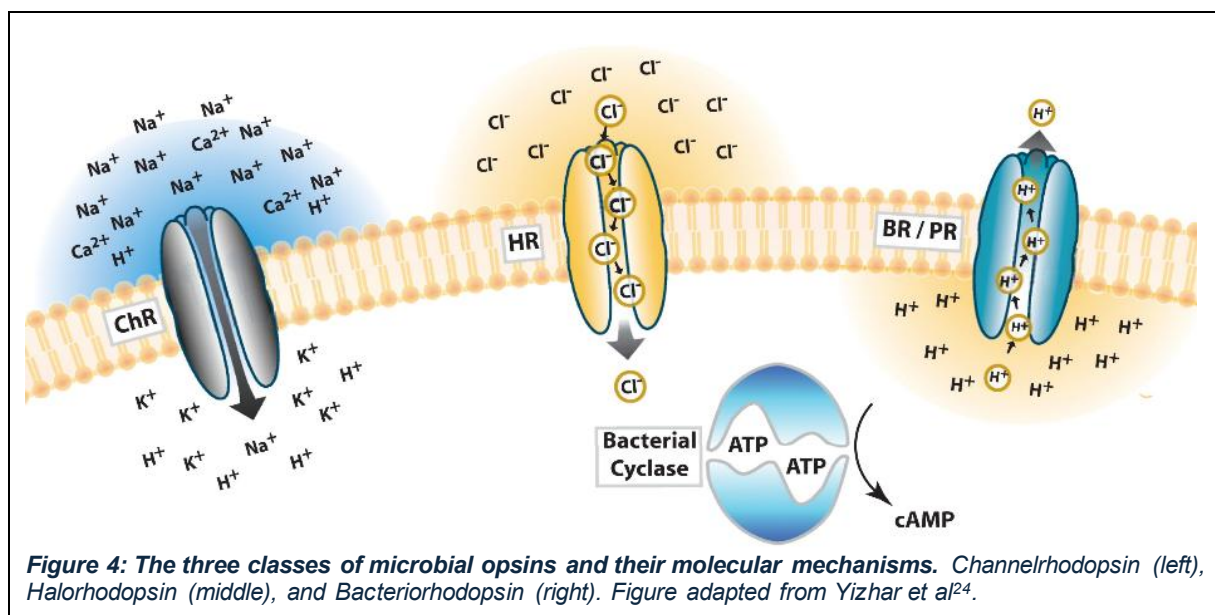
Bacteriorhodopsin was the first discovery in this field (1971), identified as a microbial, single-component light-driven ion pump. Subsequent research, spanning over the following decades, not only enhanced general understanding of bacteriorhodopsin, but also led to the discovery of numerous other members within the microbial opsin family. This family includes membrane-bound ion pumps and channels, such as halorhodopsins and channelrhodopsins, which mediate the transport of various ions across membranes in response to light.

Despite the widespread recognition of microbial opsin genes as single-component light-activated regulators of transmembrane ion conductance, it took a long time to integrate this methodology. Numerous challenges delayed their potential for optical neural control. The first challenge was the assumption that photocurrents generated by microbial opsins would be too weak and slow to effectively control neuronal activity. Another concern was the fact that microbial membrane proteins might exhibit poor expression or toxicity in fragile mammalian neurons. Lastly, there was a belief that additional cofactors such as all-trans retinal – the light-absorbing chromophore used by microbial opsins – would need to be externally supplied for experiments in intact tissues. While these assumptions initially led researchers to focus on other approaches, a breakthrough occurred in 2005. Boyden et al. demonstrated that the introduction of a single-component microbial opsin gene into mammalian neurons – without the need for additional components – enabled reliable and sustained control of millisecond-precision action potentials. Subsequent studies revealed that mature mammalian brains and all vertebrate tissues examined thus far naturally contain sufficient all-trans retinal to support a single-component approach. By 2010, the major classes of ion-conducting microbial opsins, including bacteriorhodopsin, channelrhodopsin, and halorhodopsin, had all been established as effective tools for optogenetic control in mammalian neurons²⁴.

3.3.2 Microbial Opsin Genes

In neuroscience, optogenetics demands millisecond-scale precision to match the rapid dynamics of neural processes such as action potentials and synaptic currents²⁴. Additionally, these precise manipulations must be functional within intact systems, including freely moving mammals.

The characteristics that make microbial optogenetic tools useful are rooted in the natural behaviors of their original host organisms within their ecosystems. These organisms employ seven-transmembrane proteins, encoded by type I opsin genes, to interact with and react to environmental cues crucial for survival. Type I opsins, the protein products of microbial opsin genes, are referred to as rhodopsins when bound to retinal. However, in neuroscience applications, these optogenetic tools are conveniently described as “opsins”, a term that applies to both the genes and their protein products. These proteins function as single-component light-sensing systems, where the same protein performs both light detection and ion conductance. In contrast, mammalian (type II) opsins are G protein-coupled receptors (GPCRs), consisting of opsin proteins bound to retinal, which undergo a conformational change upon absorbing photons. This activates intracellular G proteins, initiating a signaling cascade that indirectly modulates cellular activity²⁴.



The haloarchaeal proton pump bacteriorhodopsin (BR) is the first identified and most studied type I opsin (Figure 4, right). Under low-oxygen conditions, BR is abundantly expressed in the membranes of haloarchaea, where it supports an alternative energy production system by pumping protons from the cytoplasm to the extracellular medium. This process generates a proton-motive force that drives ATP synthesis²⁴.

The second category of microbial opsin genes encodes halorhodopsins (HR), which are light-driven chloride pumps initially identified in archaeobacteria (Figure 4, middle). Halorhodopsins transport chloride ions from the extracellular to the intracellular space.

The third class of microbial opsin genes are the channelrhodopsins (ChR), encoded by the green algae *Chlamydomonas reinhardtii* (Figure 4, left). They share a high degree of homology with BRs, particularly within the transmembrane helices that form the retinal-binding pocket. However, the ion-conducting activity of ChRs is largely decoupled from the photocycle. Instead, light activation opens a functional cation channel pore, making ion flux independent of retinal isomerization and rather dependent on the kinetics of channel closure. In neurons, ChR activation induces a net photocurrent dominated by cation flow along the electrochemical gradient, leading to depolarization.

In neuroscience, a widely adopted modification is applied to all new opsin genes where the original microbial codons are substituted for mammalian codons to enhance expression levels in mammalian systems.

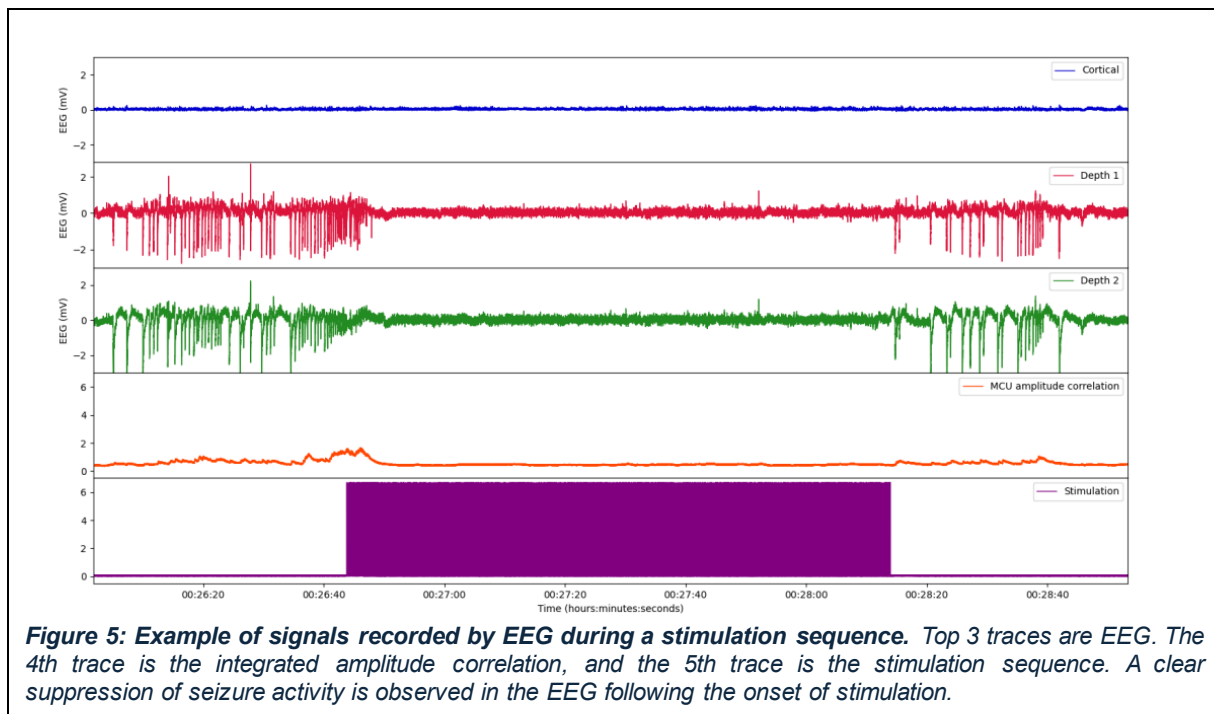
Optogenetics can have an inhibitory or excitatory effect, depending on the type of opsin. In epilepsy, which is characterized by excessive neuronal activity in the brain, inhibitory (hyperpolarizing) opsins are needed to reduce activity of excitatory neurons, or excitatory (depolarizing) opsins can be used to activate inhibitory neurons²⁴.

3.3.3 WiChR Opsin

Since other optogenetic inhibition tools have several limitations – such as high light or expression requirements – an alternative approach is the development of a potassium (K^+)-selective cation-conducting channelrhodopsin (CCR)²⁶. Such a tool would be able to hyperpolarize the cellular membrane, even in the presence of high extracellular sodium (Na^+) concentrations, by mimicking endogenous repolarization processes driven by K^+ efflux. However, the development of a K^+ -selective channelrhodopsin is challenging due to the structural dissimilarities between channelrhodopsins (ChRs) and natural K^+ channels. ChRs typically conduct protons and partially hydrated ions via a sequence of waterfilled cavities within an asymmetric pore of rhodopsin monomer. In contrast, mammalian K^+ channels have a conserved, symmetric selectivity filter with four subsequential K^+ binding sites within the axis of their tetrameric structure.

A significant advancement is the recent discovery of two natural K^+ -conducting channelrhodopsins (KCRs) in the stramenopile *Hyphochytrium catenoides* (HcKCR1 and HcKCR2)²⁶. Both channels demonstrate sufficient K^+ selectivity to inhibit the action potential firing in neuronal slices. These potassium channels feature a bacteriorhodopsin-like DTD motif in membrane helix 3 (compromising D85, T89, and D96). A subsequent search for CCRs containing this K^+ selectivity motif, led to the identification of WiChR (*Wobblia* inhibitory channelrhodopsin). WiChR is a well-expressed, light-sensitive potassium channel derived from *Wobblia lunata*. It inhibits neurons upon activation with blue light (470 nm), due to efflux of potassium and subsequent hyperpolarization of the neuron (Figure 3). WiChR exhibits enhanced K^+ selectivity, enabling more effective neuronal inhibition across a wider range of resting membrane potentials. Additionally, WiChR also has improved light sensitivity, which simplifies the light delivery for *in vivo* experiments and increases the depth for optical inhibition. Furthermore, WiChR's prolonged open-state lifetime is crucial for the pulsed stimulation protocol, which can be sustained for prolonged durations without causing heat accumulation in the tissue. WiChR has proven to have excellent inhibitory performance in neurons and cardiomyocytes.

The experiments conducted in the 4Brain lab already proved the feasibility of this opsin to suppress seizures in real-time (Figure 5)²⁷. Besides that, optical stimulation parameters were explored to maximize suppression with minimal stimulation. To limit tissue heating, the optical power is limited to 10 milliwatts in combination with a low duty cycle. In addition, different pulse frequencies and pulse durations were tested, where it was found that lower frequency stimulation (5 Hz, 5 ms) is less effective to suppress seizures compared to high frequency stimulation (10 Hz, 2.5 ms) despite the identical duty cycle.



3.3.4 Delivery Systems

As previously mentioned, optogenetics necessitates the delivery of gene constructs into target cells, which requires a delivery system that is both safe and efficient for *in vivo* applications. While non-viral delivery of genetic material is extensively studied and even therapeutically used, viral vectors remain the predominant choice for *in vivo* gene delivery²⁸. Viral vectors are considered the gold standard and have been used extensively due to their high transduction efficiency and overall safety profile²⁹. These vectors are derived from genetically modified viruses, that have been engineered to be replication-deficient, thereby preventing viral propagation within the host³⁰.

There are several types of viral vectors designed to introduce new genetic material into the nucleus of host cells. These vectors can be classified based on their serotypes, which refer to subgroups characterized by distinct surface proteins that confer specific tissue tropisms. Some vectors are able to integrate their genetic material into the host genome. Retroviral vectors, including the subclass of lentiviral vectors, carry genetic elements encoding reverse transcriptase and integration enzymes (pol gene). Lentiviral vectors offer the advantage of inserting long transgenes (up to ~8kb) and ensure long-term expression in both dividing and non-dividing cells, such as CD34+ and neuronal cells. However, the principal drawback of integrating vectors is the increased risk of insertional mutagenesis and oncogenesis associated with genome integration²⁹. Therefore, non-integrating vectors may be more interesting as they minimize the risk of oncogenesis. For example, adenoviral vectors demonstrate preferential tropism for CNS cell types, making them suitable candidates for CNS gene delivery. Nevertheless, their clinical application remains limited by significant toxicity and immunogenicity³¹.

In neuroscience, adeno-associated virus (AAV) vectors are most frequently used and are widely regarded as the ideal vector for the CNS. Their key advantages include the absence of natural genomic integration and minimal immunogenicity, while still achieving efficient transduction of both dividing and non-dividing cells. AAVs are single-stranded DNA viruses with a genome composed of a Cap and Rep Open Reading Frame (ORF), flanked by palindromic nucleotide sequences, known as inverted terminal repeats (ITR). Cap encodes structural proteins and an assembly-activating protein, whereas Rep encodes proteins essential for transcription. ITRs cause the genetic material to exist in a hairpin structure, allowing synthesis of the second DNA strand via host cell DNA polymerase. Multiple AAV serotypes exist and

exhibit distinct tissue tropisms, which can be further modified through genetic engineering to create recombinant AAVs (rAAV)³².

Although AAVs do not actively integrate into the host genome, their genetic material can occasionally integrate at sites of double strand breaks or at the human AAVS1 locus via host-mediated mechanisms. By removing the Cap and Rep genes, AAV integration into the human AAVS1 gene is prevented, enhancing the safety profile relative to other vectors³².

The gene expression cassette in rAAVs typically consists of a promoter – a regulatory DNA sequence that initiates transcription – the transgene of interest, and a transcription termination signal, all flanked by the ITRs³¹. To enhance vector performance, AAVs are often pseudotyped by combining elements from different serotypes. This study uses AAV2/9, a recombinant AAV incorporating AAV2 ITRs and an AAV9 capsid, known for its broad tissue tropism and ability to cross the blood-brain barrier^{33, 34}. Moreover, AAV9 has been shown to undergo axonal transport, resulting in distributed expression³⁵. A major limitation of AAV vectors, however, is their restricted packaging capacity, which limits the size of the expression cassette to 4.7 kilobases (kb)³². As a result, large cell type-specific promoters could not be used in the present study. For instance, the Ca²⁺/calmodulin-dependent kinase II alpha (CaMKII α) promoter, which drives expression in excitatory neurons, is approximately 1.3 kb in length. In contrast, the pan-neuronal human synapsin I (hSYN) promoter is considerably smaller at 499 base pairs (bp), making it a more feasible choice within the AAV vector constraints³⁶.

The vector used in this study is AAV2/9.hsyn-WiChR1_ts_mscarlet_er-WPRE (Figure 6)²⁷. This construct is based on an AAV2/9 backbone, incorporating the WiChR1 transcript, under control of the human synapsin promoter (hSYN) to promote neuron-specific expression²⁶. The WiChR1 coding sequence is fused to a Kir2.1 membrane-targeting sequence (ts) to facilitate Golgi export and membrane targeting, as well as to mScarlet (mscarlet_er), a red fluorescent protein for visualization purposes. Additionally, the vector includes the Woodchuck Hepatitis Virus Posttranscriptional Regulatory Element (WPRE), which improves mRNA stability and protein expression³⁷.

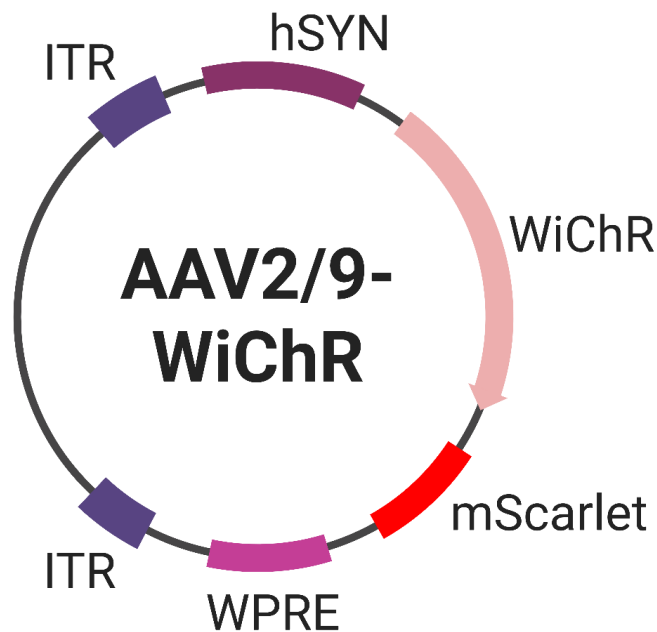


Figure 6: Schematic representation of the AAV2/9.hsyn-WiChR1_ts_mscarlet_er-WPRE vector. The expression cassette is flanked by ITRs and includes the hSYN and WPRE for neuron specificity and mRNA stability, respectively. The central component is the WiChR1 transgene, with a targeting sequence for membrane targeting and the fluorescent mScarlet protein sequence for visualization.

3.4 Neurotoxicity

Neurotoxicology is a field that studies the changes of function and structure of the nervous system following exposure to chemical, biologic, or physical agents. This covers environmental factors that may alter the nervous system, including life-style and food, but also exposure to pharmaceuticals and biologics, such as opsin proteins. Even though optogenetic tools like WiChR have become essential for studying and modulating neural circuits, it is essential that the presence of these proteins does not itself interfere with normal circuit function and structure³⁸.

Assessment of neurotoxicity typically involves histopathological analysis of brain tissue to detect structural damage and inflammatory markers. Since this study targets the hippocampus, measuring hippocampal volume is a logical first step in this process. This is based on a study by Goossens et al., where neuronal loss was investigated through measuring hippocampal volume via T2-weighted MR images³⁹. As the current study employed immunohistochemistry, the hippocampal volume metric was replaced by hippocampal surface area. Other studies propose an investigation of the cell layer thickness and/or cell count of the pyramidal layers and the hilus of the dentate gyrus³⁹⁻⁴¹.

In addition to structural analysis, histopathology provides insights into inflammation, which is a common response to both gene therapies and implantations^{41, 42}. The inflammatory response linked to neuronal death is strongly modulated by microglia, the brain's resident immune cells⁴³. They regulate neuronal and synaptic activity, maintain homeostasis, and clear debris. When activated by neuronal death or foreign material, they adopt an inflammatory state and migrate to the affected site^{43, 44}. They may promote an inflammatory response and tissue repair by producing pro-inflammatory factors, such as cytokines and growth factors⁴³. While they have a protective function, excessive or prolonged activation can contribute to neuronal damage and degeneration⁴². Therefore, minimizing inflammation is crucial when evaluating different WiChR concentrations. Activated microglia are typically characterized through the marker ionized calcium binding adaptor molecule 1 (Iba1), though other markers like Cox-2, TNF- α , IL-1 β , and IL-6 are also used to assess their inflammatory state^{41, 44}.

Neurotoxicity in optogenetic applications can arise from two main sources: the viral delivery system (AAV) or the expressed transgene (opsin). Strong expression of opsins has been shown to increase the risk of structural abnormalities and potential cell death. Fujita et al. showed that excess expression of proteins may induce disturbances and potential cell death due to expression limits⁴⁵. Furthermore, Miyashita et al. demonstrated that long-term, high level expression of ChR2 can induce the formation of elongated cylinders and large, round, calyx-like structures, termed "ChR2 puncta", on axons in rats³⁸. High opsin expression can increase membrane capacitance and, in some cases, disrupt membrane integrity, potentially leading to cell death as well⁴⁶. The choice of promoter plays a crucial role here. Strong promoters that drive robust and prolonged expression are often necessary for effective optical control of neurons. However, use of strong promoters, such as α CaMKII and hSYN, for extended periods increases the risk of structural abnormalities and potential cell death³⁸.

Opsin expression can also trigger immune responses. Maimon et al. showed that ChR2 can initiate a cytotoxic T lymphocyte-mediated immune response, directly leading to loss of optogenetic activity in the central and peripheral nervous systems, neuronal death, and muscle atrophy⁴⁶. Immunosuppressive treatment extended opsin expression in wild-type rats, further supporting the role of the immune system.

Separate from the opsins itself, immune responses to AAV delivery systems have been reported. While AAVs are generally considered safe and are widely used in gene therapy, according to Pérez et al., their immunogenicity in the CNS remains under-investigated³⁵. Reported reactions include the development of neutralizing antibodies in human trials, inflammation in response to AAV delivery into the spinal cord in mice, and immune reactions in optogenetic contexts⁴⁶. Other studies report white and gray matter necrosis along the injection tract³⁵. Various approaches have been suggested to reduce these responses, such

as immunosuppressive drugs, careful dosing, use of decoy capsids, engineered AAV variants, or use of non-viral delivery methods. However, some studies do not show evidence of AAV-related neuroinflammation, indicating a strong dependence on the experimental protocol.

3.5 Research Objectives

Pilot experiments demonstrated effective optogenetic suppression of seizures in the IHKA mouse model of epilepsy using the WiChR opsin, indicating its therapeutic potential. However, in first follow-up experiments, these initial results could not be reproduced due to severe hippocampal degradation, indicating potential toxicity of the WiChR opsin. This observation prompted the current study, which aims to assess both the safety profile of the WiChR opsin and its efficacy in seizure suppression using a responsive neuromodulation (RNM) system.

In the first part of this thesis, the following research question is addressed: *“Is there a dilution of the WiChR viral vector that does not induce toxicity in the mouse hippocampus?”*. To answer this, the toxicity of the WiChR opsin is assessed in the hippocampus of healthy mice. Three viral dilutions (1/5, 1/10, or 1/100) are tested, and toxicity is evaluated using immunohistochemistry. It is hypothesized that lower concentrations of WiChR will result in reduced toxicity. To further assess WiChR’s safety, the non-toxic WiChR dose is injected into the IHKA mouse model following epilepsy induction. Based on previous successful transfections in this model, described in the paper by Lasure et al., it is hypothesized that the addition of WiChR in the IHKA mouse model will be well tolerated²⁷.

The second part of this thesis explores the following research question: *“Does optogenetic stimulation using a non-toxic dilution of WiChR effectively suppress seizures in the KA mouse model of TLE?”*. To investigate this, KA animals are injected with WiChR and implanted with an optrode. The mice are then connected to the RNM system to assess whether seizure duration is affected by optogenetic stimulation. It is hypothesized that illumination of WiChR will reduce seizure duration, due to its inhibitory effect as a light-sensitive potassium channel.

4 Materials And Methods

4.1 Experimental Design

This study comprises two main parts: (1) A toxicity study, which includes a dose-titration experiment and a WiChR-KA experiment, and (2) a responsive neuromodulation experiment (Table 3).

In the dose-titration experiment, animals were injected with different doses of the WiChR opsin (1/5, 1/10, or 1/100) in the left hippocampus, while the right hippocampus was injected with saline as a control. Approximately seven weeks post-injection, the histology commenced, beginning with transcardial perfusion, followed by cryosectioning and immunohistochemistry.

For the WiChR-KA experiment, animals first received a right intrahippocampal injection of kainic acid (KA). After a three-week waiting period, the non-toxic WiChR dose was injected. One animal, injected with KA and saline, served as a control. Histology was performed three weeks later.

In the responsive neuromodulation experiment, KA was injected into the right hippocampus, and the non-toxic WiChR dose was introduced three weeks thereafter. Two weeks post-WiChR injection, animals underwent implantation of the optrode and scalp electrodes. After one week of recovery, EEG recordings were initiated, followed by histology.

Experiment	Protocol
Dose-titration (Batch 1)	WiChR → IHC
WiChR-KA (Batch 2)	KA → WiChR → IHC
Responsive neuromodulation (Batch 3)	KA → WiChR → Optrode → EEG → IHC

Table 3: Experimental design. In the dose-titration experiment, healthy mice received an intrahippocampal injection of the WiChR opsin, followed by perfusion and histology at least three weeks later. In the WiChR-KA experiment, mice were first injected with KA, followed by WiChR three weeks later, and perfused at least one week after that. In the responsive neuromodulation experiment, KA was injected first, followed by WiChR three weeks later. After two weeks, optrode and scalp electrode implantation was performed. Recordings began after a one-week recovery period. All animals were subsequently perfused for histological analysis.

4.2 Animals

Twenty-eight C57BL/6J mice, aged 6 to 8 weeks upon arrival, were acquired from Janvier. The animals were delivered in separate batches of 9, 6 and 13 mice, depending on the running experiment at the time. Upon arrival, animals underwent a one-week acclimatization period before the start of experimental procedures. Animals were treated according to the European guidelines (directive 2010/63/EU). All experiments described in the following were approved by the Animal Experimental Ethical Committee of Ghent University (ECD 22-39) and complied with the ARRIVE guidelines. Nine mice were used for the dose-titration experiment (Batch 1), six for the WiChR-KA experiment (Batch 2), and ten for the responsive neuromodulation experiment (Batch 3). Batches 1 and 2 consisted entirely of male mice, while Batch 3 had an equal male-to-female ratio (50:50). In addition, a follow-up experiment included an injector comparison (n=1, male) and a second responsive neuromodulation test with KA-pretreated animals (n=2, female). Before surgeries, they were housed together in open cages. After surgeries and during recordings, they were housed individually in open cages. All cages were placed in a room with a 12/12-hour light/dark cycle and maintained at a controlled environmental temperature (20-24°C) and humidity (40-60%). Food and water were provided at libitum.

4.3 Kainic Acid and WiChR Injections

In all experiments except for the dose-titration study, the first step was to create the mouse model for TLE. This was accomplished via injection of kainic acid into the right dorsal hippocampus to induce status epilepticus and the occurrence of spontaneous recurrent seizures.

Mice were anesthetized using isoflurane (induction 5%, maintenance 2%) in oxygen (induction 1 L/min, maintenance 0.8 L/min) and secured in a stereotaxic frame (Neurostar Robot Stereotaxic system, Neurostar, Tübingen, Germany) using ear bars. 50 nL kainic acid (200 ng kainic acid in 50 nL saline) was unilaterally injected (Nanoliter 2020, WPI, Florida, US) into the dorsal hippocampus with a flow rate of 250 nL/min at the stereotaxic coordinates -2.0 mm anterior-posterior (AP) relative to bregma; +1.5 mm medial-lateral (ML) relative to bregma; -2.0 mm dorsoventral (DV) relative to dura (Figure 7, red).

Three weeks thereafter, WiChR was injected in all three batches, following the same protocol. 1 μ L of the AAV2/9.hsyn-WiChR1_ts_mscarlet_er-WPRE vector was injected with a flow rate of 100 nL/min to obtain expression of the WiChR opsin, but different concentrations, injectors, and coordinates were used (Table 4).

Regarding Batch 1 (n=9), the original WiChR concentration (titer $8.47\text{E}+12$ vg/mL) was diluted to 1/5, 1/10, or 1/100 of its original concentration, with three animals per dilution. The vector was injected (Nanoliter 2020, WPI, Florida, US) into the left dorsal hippocampus at the stereotaxic coordinates AP -2.0 mm relative to bregma; ML -1.5 mm relative to bregma; DV -1.8 mm relative to dura (Figure 7, blue). Saline was injected into the contralateral hippocampus at the same coordinates to serve as a control.

In Batch 2 (n=3), two animals were injected (Nanoliter 2020, WPI, Florida, US) with 1/100 WiChR at the stereotaxic coordinates AP -2.0 mm relative to bregma; ML +1.5 mm relative to bregma; DV -2.0 mm relative to dura (Figure 7, red). One control animal received KA and saline in the right dorsal hippocampus but was not injected with WiChR.

In Batch 3 (n=13), the first 10 animals were injected with 1/100 WiChR in the right hippocampus at the stereotaxic coordinates AP -2.0 mm relative to bregma; ML +1.5 mm relative to bregma; DV -2.0 mm relative to dura (Figure 7, red), using a Hamilton syringe (Hamilton, Neuros syringe, 65460-03, Hamilton Company). Following this experiment, one animal was repurposed to compare the Hamilton and Nanoliter injector. Additionally, two extra KA animals – originally part of another experiment – were injected with the same WiChR dilution, but using a Nanoliter injector (Nanoliter 2020, WPI, Florida, US). For the repurposed animals, injections were performed using the original stereotaxic coordinates: AP -2.0 mm, ML +1.5 mm, and DV -1.8 mm, all relative to the bregma.

Dorsal hippocampus

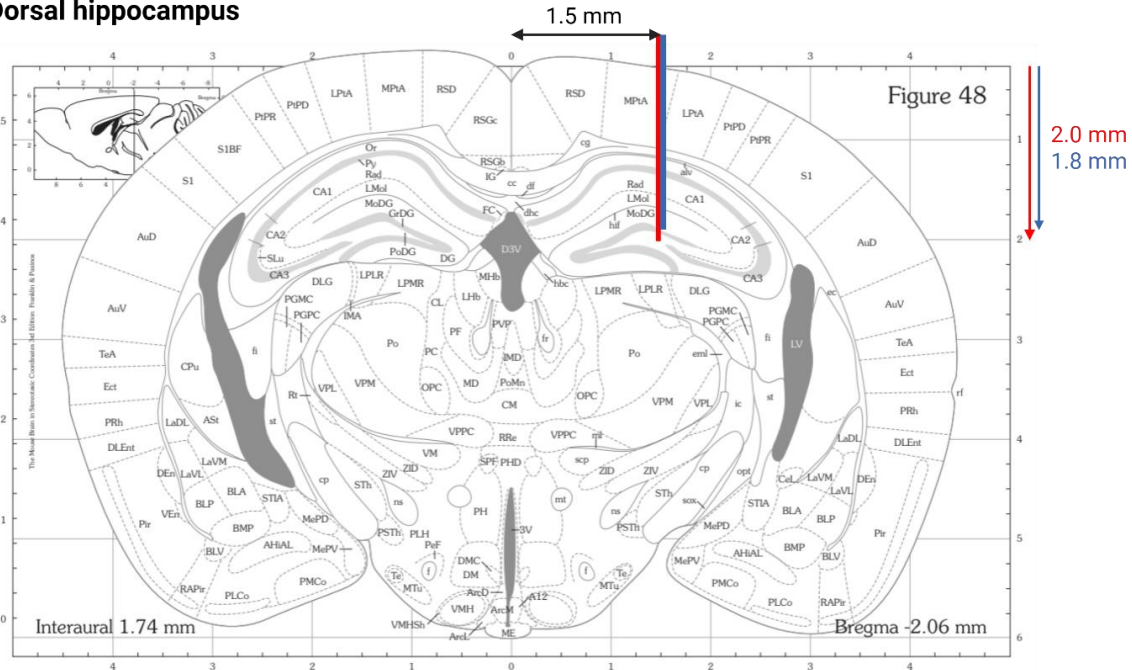


Figure 7: Stereotaxic coordinates of kainic acid injection, vector injection, and electrode implantation. The injection site of kainic acid (red) in the dorsal hippocampus at -2.0 mm AP-bregma, +1.5 mm ML-bregma, -2.0 mm DV-dura. Injection of WiChR vector at the same coordinates (red) for all batches, except Batch 1 and 3, which were injected higher at -2.0 mm AP-bregma, +1.5 mm ML-bregma, -1.8 mm DV-dura (blue). This figure shows sections at -2.06 mm relative to bregma, but the injections and implantations were at -2.0 mm and -1.8 mm relative to bregma. (Figure from Franklin et al. and adapted with BioRender⁴⁷)

Experiment	Injected components	Coordinates (mm)	Injector
Dose-titration MVT1-MVT9	MVT1-MVT3: 1/5 WiChR / Saline MVT4-MVT6: 1/10 WiChR / Saline MVT7-MVT9: 1/100 WiChR / Saline	-2.0 AP – bregma +/-1.5 ML – bregma -1.8 DV – dura	Nanoliter
WiChR-KA SL043, SL044, SL046	SL043: KA + saline SL044, SL046: KA + WiChR	-2.0 AP – bregma +1.5 ML – bregma -2.0 DV – dura	Nanoliter
Responsive neuromodulation SL051-SL060, EL034, EL037	KA + WiChR	-2.0 AP – bregma +1.5 ML – bregma -2.0 DV – dura	SL: Hamilton EL: Nanoliter

Table 4: Overview of experimental groups, coordinates and injectors per experiment. The symbol “/” is used to indicate that the components were injected into the ipsilateral and contralateral hippocampus, while the symbol “+” denotes that both components were injected into the same hippocampus. Note: The EL animals of the responsive neuromodulation experiment were injected at a higher DV coordinate of -1.8 mm relative to the bregma.

4.4 Histology

Neurotoxicity and/or expression were assessed via immunohistochemistry. Mice were perfused at least three weeks after the injection of the viral vector. Animals were anesthetized using a mixture of isoflurane (5%) and oxygen (1 L/min), followed by an intraperitoneal overdose of sodium pentobarbital (Dolethal, 200 mg/kg, Vetoquinol) (0.35 – 0.40 mL). After heparin injection (0.03 mL) into the heart, transcardial perfusion was performed with phosphate buffered saline (PBS), followed by 4% paraformaldehyde (PFA). Brains were isolated and postfixed in 4% PFA for 24 hours, after which it was cryoprotected in a 30% sucrose solution. Subsequently, the tissue was snapfrozen in isopentane cooled with liquid nitrogen (-196°C), and stored at -20 °C until cryosectioning the next day.

Coronal sections (40 µm) of the brain were made at the level of the hippocampus with a cryostat (Leica CM1950) and stored in antifreeze medium at -20°C. The following day, immunofluorescence staining was performed on six to eight slices per brain, ranging from -1.06 mm to -2.30 mm AP relative to bregma, with an interslice distance of ~0.12 mm.

Slices were first incubated in H₂O₂ to block endogenous peroxidase activity. This was followed by incubation in blocking buffer containing PBS, 0.4% Fish Skin Gelatin (40-50% in H₂O), and 0.2% Triton X to saturate non-specific binding sites and reduce background staining. Sections were then incubated overnight at 4°C with primary antibodies diluted in blocking buffer (1:1000) (Table 5). The next day, sections were incubated with secondary antibodies diluted in blocking buffer (1:1000) (Table 5), followed by nuclear DAPI staining (1 µg/mL in PBS). Finally, slices were mounted on microscope slides with Vectashield Mounting Medium Fluorescence covered by a coverslip. The slides were stored at 4°C until imaging. Slides were digitized using Panoramic 250, 3D Histech, 40×magnification at HIRUZ (<https://hiruz.be/>) and uploaded on the slide viewing application CaseViewer.

Staining	Primary antibody	Secondary antibody	Fluorescent Color
WiChR	Chicken anti-RFP	Goat anti chicken	Red
Microglia	Rabbit anti-Iba1	Goat anti rabbit	Green

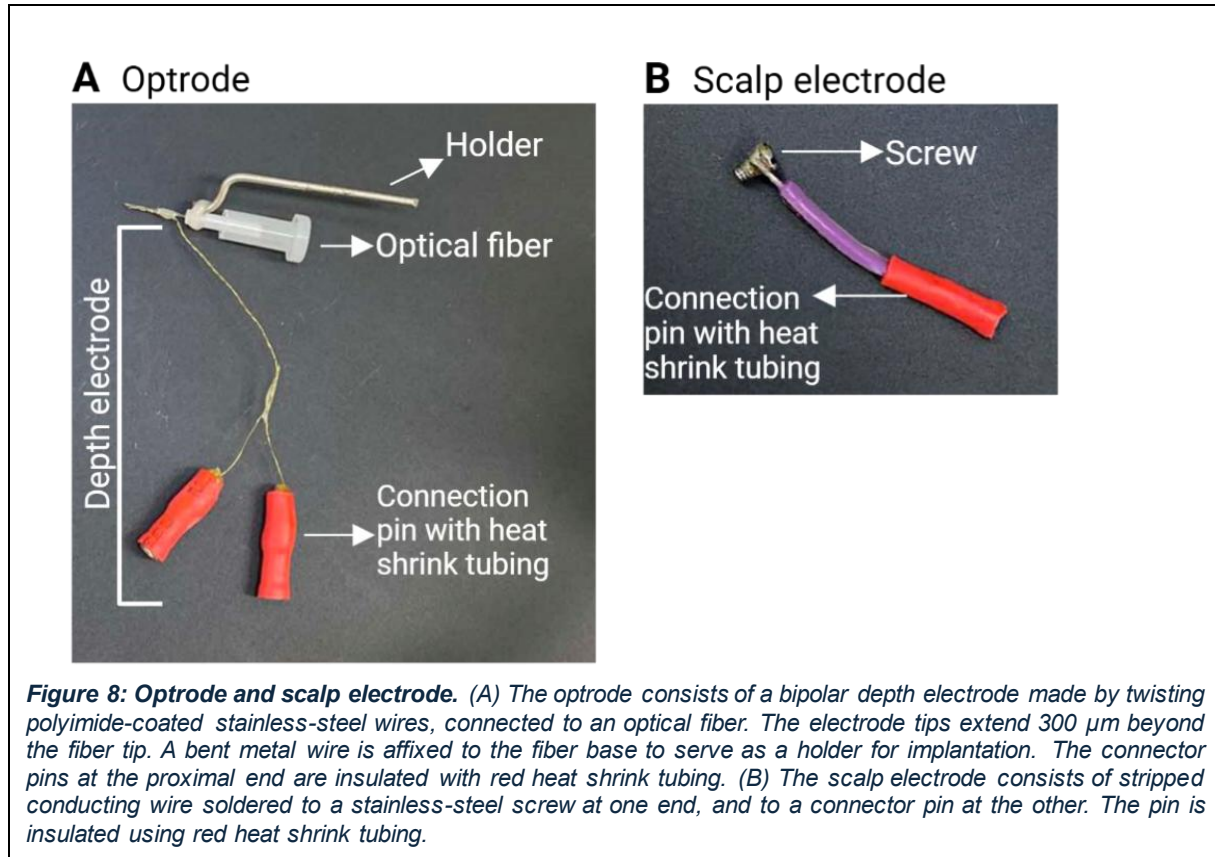
Table 5: Overview of primary and secondary antibodies used in immunohistochemistry staining.

4.5 Optrode Fabrication

In the responsive neuromodulation experiment, the animals' electrical brain activity was monitored using electroencephalography (EEG). This was done by implanting each animal with an optrode and two scalp electrodes. The optrode was implanted in the left hippocampus and consisted of a bipolar depth electrode that records the EEG, and an optical fiber that illuminates the hippocampus in case of seizure detection. In addition, two scalp electrodes were implanted contralaterally as ground/reference electrodes.

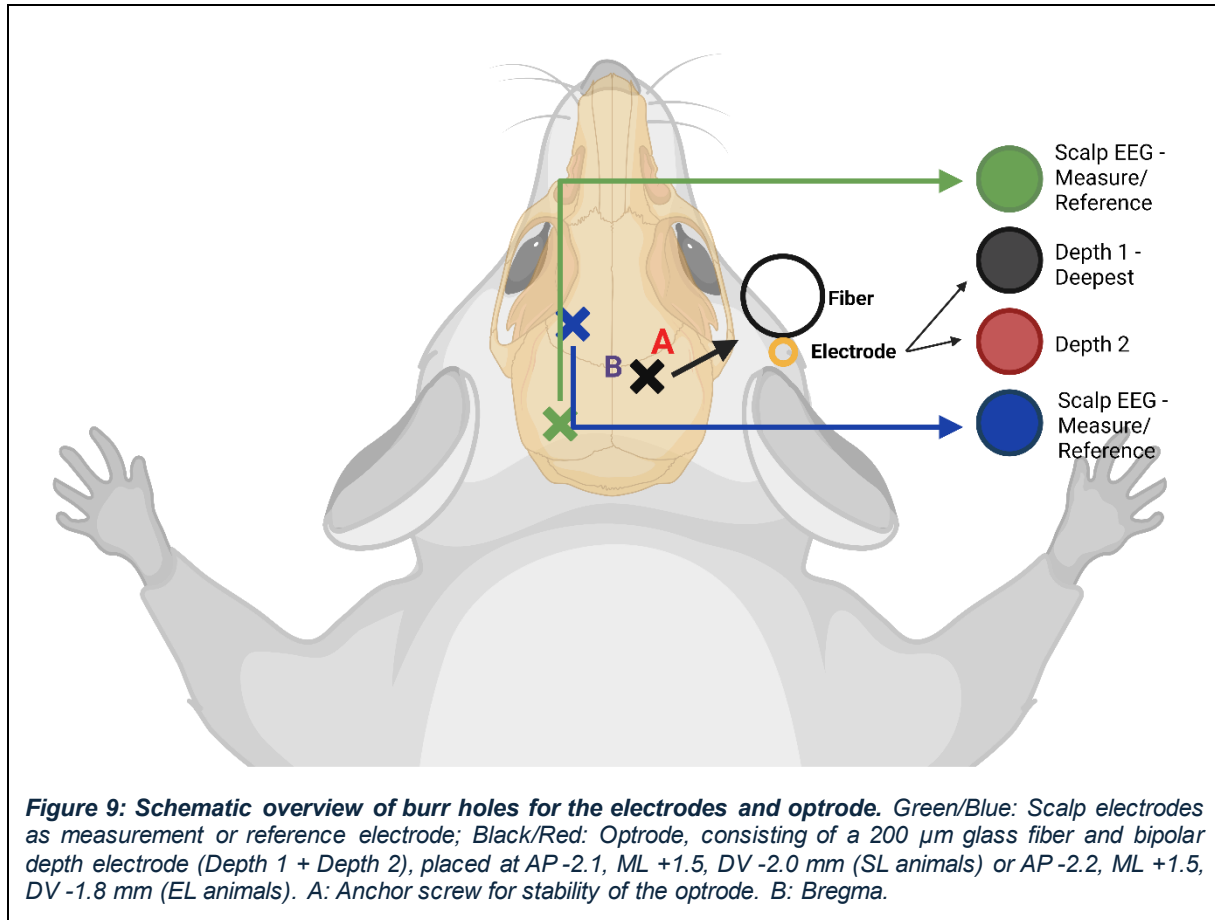
The depth electrode was made by twisting ~15 cm of polyimide-coated stainless-steel wires (ø70 µm, California Fine Wire, USA). After cutting open the loop, the polyimide coating at both ends was removed and clamped between two connector pins which were then soldered together. The connector pins were surrounded by heat shrink tubing (2.4 mm, RS Components, UK) for insulation and the electrode tip was cut diagonally to a length of ~3 cm. The depth electrode was then connected to the optical fiber (ø200 µm, length 5 mm, 1.25 mm ferrule, R-FOC-L200C-22NA, RWD, USA) using UV cement (Dental Products, Kemdent, UK) under a light microscope. Care was taken to ensure that the electrode tips extended 300 µm beyond the tip of the optical fiber, enabling effective illumination of the transduced neurons. A bent metal holder was attached to the base of the optical fiber via UV cement to ease the implantation procedure (Example of optrode: Figure 8, A).

Scalp electrodes were made by stripping ~0.5 cm from both ends of approximately 3 cm conducting wire (RS components, UK). One end was hooked around a stainless-steel screw ($\varnothing 1.57$ mm, Bilaney Consultants, Germany) and soldered in place, ensuring the wire sat as close to the screw head as possible to leave enough shaft length for surgical insertion. Afterwards, a connector pin was soldered to the other end of the wire and covered with heat shrink tubing (Example of scalp electrode: Figure 8, B).



4.6 Electrode and Optrode Implantation

Implantations for Batch 3 were performed two weeks after the injection of the WiChR opsin. The preparation of the animals followed a similar protocol as described for the injection procedures, with isoflurane maintained at 2%. Three initial burr holes were drilled to secure one measure scalp electrode, one reference scalp electrode, and an anchor screw positioned near the intended optrode coordinates (Figure 9). Then, a burr hole was made for optrode implantation. For the first 10 animals, the optrode was implanted at AP -2.1 mm relative to bregma, ML +1.5 mm relative to bregma, and DV -2.0 mm relative to dura, with the deepest point being the electrode tip. For the two repurposed animals, the coordinates were AP -2.2 mm, ML +1.5 mm, and DV -1.8 mm, all relative to bregma. The optrode was secured using a micro-screw and fixed to the anchor screw. The electrode and optrode wires were connected to four connector pins, and the assembly was secured in place, leaving the fiber cap accessible.



4.7 Recording set-up

The mice of Batch 3 were connected to an adaptable responsive neuromodulation system – the minimalist embedded system – designed by Lasure et al.²⁷. The system consists of a “regular” recording set-up consisting of a pre-amplifier, an amplifier with high pass filter (x500, 2nd order HPF 0.159Hz) and a National Instruments Digital Acquisition Card (NIDAQ) to digitize the signal for storage and analysis. The signals were recorded via the custom Neuron software, which runs on MATLAB® R2007a (The MathWorks, Inc., USA). Within this set-up, an embedded system for responsive neuromodulation is integrated – as described in more detail in the paper by Lasure et al. – which records the amplified and filtered EEG-signals²⁷. The embedded system contains an MSP430 microcontroller which computes the amplitude correlation metric. Seizures are detected when the amplitude correlation metric crosses the detection threshold. Seizures are considered finished when the amplitude correlation metric crosses a lower threshold to ensure a seizure is fully finished. An adjustment compared to the detection algorithm as described in Lasure et al. was made to allow adaptive thresholding²⁷. A running average was made of the amplitude correlation metric as described in the equation below. The detection threshold was then defined as a “ $factor_{up}$ ” x the adaptive amplitude correlation. Through this implementation, the detection thresholds adjust automatically to the strength of the EEG signals of the animal under consideration, removing the need to manually adjust the thresholds as in the previous experiments.

$$AC_{adaptive}[n] = gain_{adaptive} * AC_{adaptive}[n - 1] + (1 - gain_{adaptive})AC[n]$$

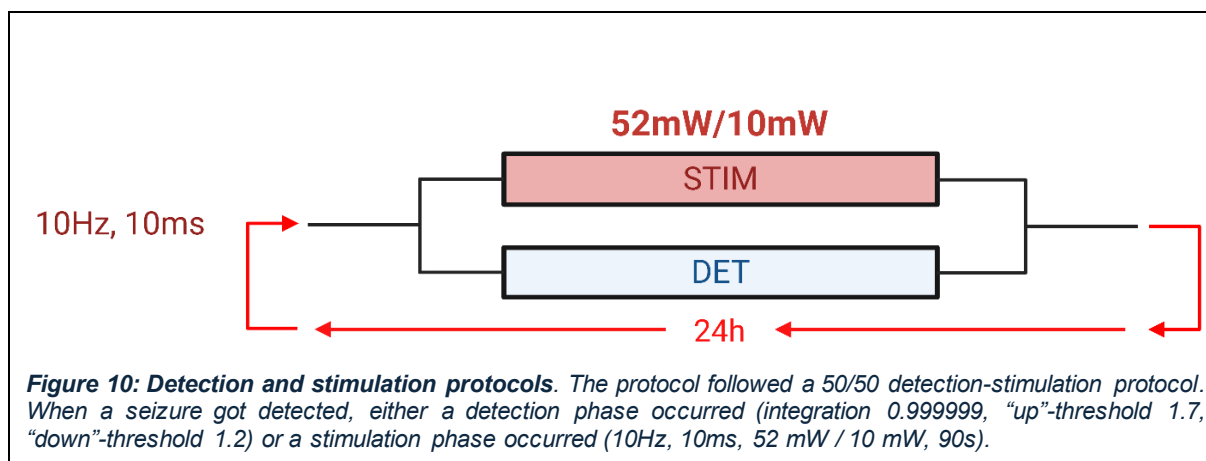
$$Threshold_{up} = factor_{up} * AC_{adaptive}[n]$$

$$Threshold_{down} = factor_{down} * AC_{adaptive}[n]$$

The embedded system drives the optical stimulation source, a 470 nm laser (Arktis 470nm Laser), with TTL-pulses allowing control of pulse frequency, pulse duration and stimulation duration. Stimulation power is configured by measuring the optical output power at the end of the patch cord with a photometer and adjusting the source to deliver the desired output power. The optical pulses are delivered to the animals through an optical patch cord. Both the optical patch cord and electrical cable run through a swivel to allow free movement of the animals, allowing long-term recording with a minimal burden on the mice.

The microcontroller controls the stimulation protocol. In the software, various stimulation “days” can be programmed, with each day consisting of various detection and stimulation elements. In this study, a 50/50 detection–stimulation protocol was developed, meaning that either a detection or stimulation element was randomly executed for each detected seizure, with an equal distribution between the two over the session (Figure 10). For the detection parameters, an adaptive gain of 0.999999 was selected, an up-factor of 1.7, and a down-factor of 1.2. The stimulation phase occurred when a seizure got detected, and lasted for 90 seconds at 10 Hz, with a pulse duration of 10 ms. The power was different for the SL animals and EL animals: In SL animals, a maximum power of 52 mW was used, while in EL animals, the parameters were based on those reported by Vierock et al., who employed a power setting of 10 mW in their experiments²⁶. Recordings lasted for 24 hours, of which 8 hours were analyzed.

It is important to note that brief spontaneous seizure events may have coincided with stimulation onset, but terminated naturally, creating the false impression of a therapeutic effect (Addendum, Figure I). To mitigate this, the detection threshold was relatively high (factor of 1.7) to reduce false positives.



4.8 Data Analysis

All statistical analyses were performed in R (version 4.4.2), and the packages *ggplot2* and *ggpubr* were used to generate boxplots.

4.8.1 Neurotoxicity

For the toxicity study, image analysis was performed on three brain slices: one at the injection point and two adjacent slices located 120 μ m anterior and posterior to it. Image processing and analysis were conducted using the FIJI (ImageJ) software package. After exporting the histological images (Raw image format; 1:8) as a Tiled TIFF file from CaseViewer, images were split into three channels representing the cell nuclei (DAPI/Blue), microglia (FTIC/Green), and WiChR (TRITC/Red).

Toxicity was assessed based on two variables: hippocampal degeneration and amount of microglia. Hippocampal degeneration was measured by manually delineating both hippocampi three times on the DAPI-stained images (Figure 11). These triplicates were averaged to obtain a single value per slice. The ratio of the injected hippocampus relative to the contralateral hippocampus was calculated (Formula below).

$$\text{Hippocampal degeneration} = \frac{\text{Surface area Injected hippocampus}}{\text{Surface area Contralateral hippocampus}}$$

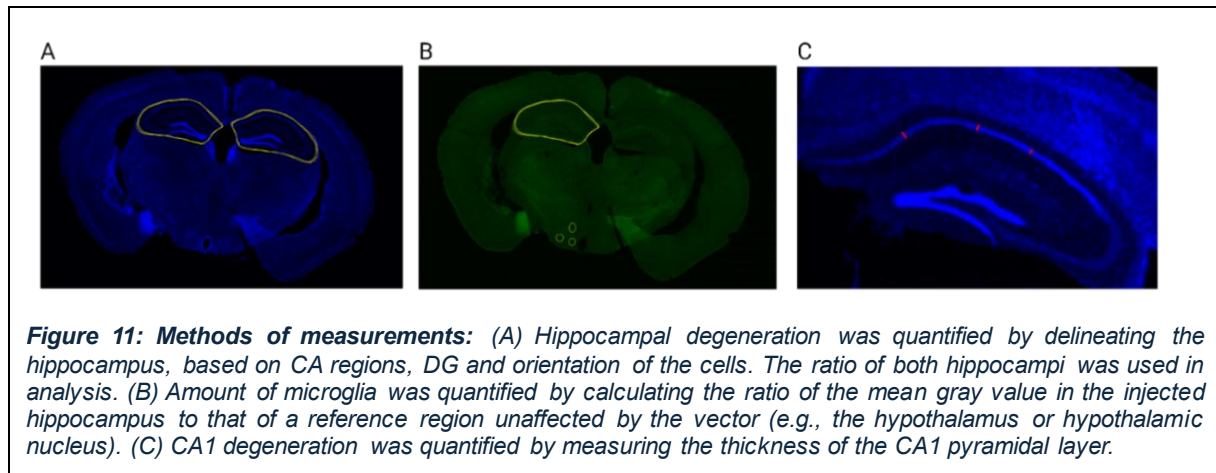
To measure the amount of microglia based on intensity, the FITC/Green channel was used. The mean grey value was taken of the injected hippocampus, where the same ROIs were used as for measuring hippocampal degeneration, also averaged based on three measurements. These values were corrected for intensity differences between slices by measuring the mean grey value of a reference region in the hypothalamus, also in triplicate (Figure 11; Formula below).

$$\text{Microglia abundance} = \frac{\text{Mean Grey Value Injected Hippocampus}}{\text{Mean Grey Value Reference}}$$

For each animal, the average of the three slices around the injection point was used as a single datapoint for statistical analysis. Due to limited data points and the absence of normality, the non-parametric Kruskal-Wallis test was performed for both surface and microglia to evaluate the effect of WiChR dilution on hippocampal toxicity between the three independent groups. A significance level of 0.05 was implemented.

To further confirm the non-toxicity of the seemingly safe 1/100 dilution, the thickness of the CA1 region was manually measured at three distinct locations and averaged to obtain a single value per slice (Figure 11). To enable comparison across animals, the ratio of the injected hippocampus as opposed to the contralateral hippocampus was taken again (Formula below).

$$\text{CA1 degeneration} = \frac{\text{Layer thickness Injected hippocampus}}{\text{Layer thickness Contralateral hippocampus}}$$



4.8.2 Responsive neuromodulation

For the responsive neuromodulation experiment, all seizures were manually annotated using the Neuron software in a blind manner, i.e. without knowledge of the stimulation or detection information from the embedded system. Seizures were annotated as EEG segments containing repetitive spiking with significantly higher amplitude than baseline, occurring at a frequency of at least 2 Hz for a minimum duration of 7 seconds, as described in Lasure et al. (Figure 12)²⁷.

To evaluate the effectiveness of the responsive neuromodulation system, EEG data were processed using Python (version 3.12.10) to classify seizures as either stimulated (illumination) or non-stimulated (no illumination) for each animal. Statistical comparison of seizure durations under these two conditions within the same animal were performed using the non-parametric Wilcoxon signed-rank test, as no normality was observed, with a significance threshold level of 0.05.

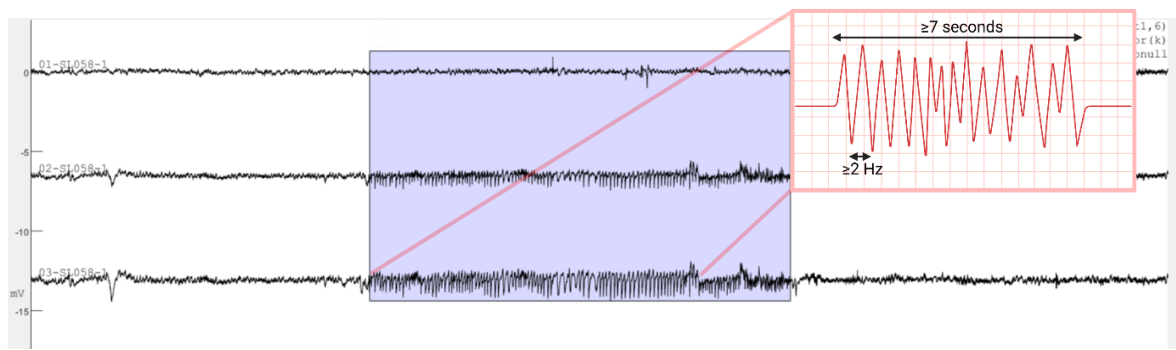


Figure 12: Seizure annotation. A seizure is defined as a spike-wave discharge with a frequency of at least 2 Hz and a duration of at least 7 seconds.

5 Results

5.1 Neurotoxicity

5.1.1 Dose-Titration

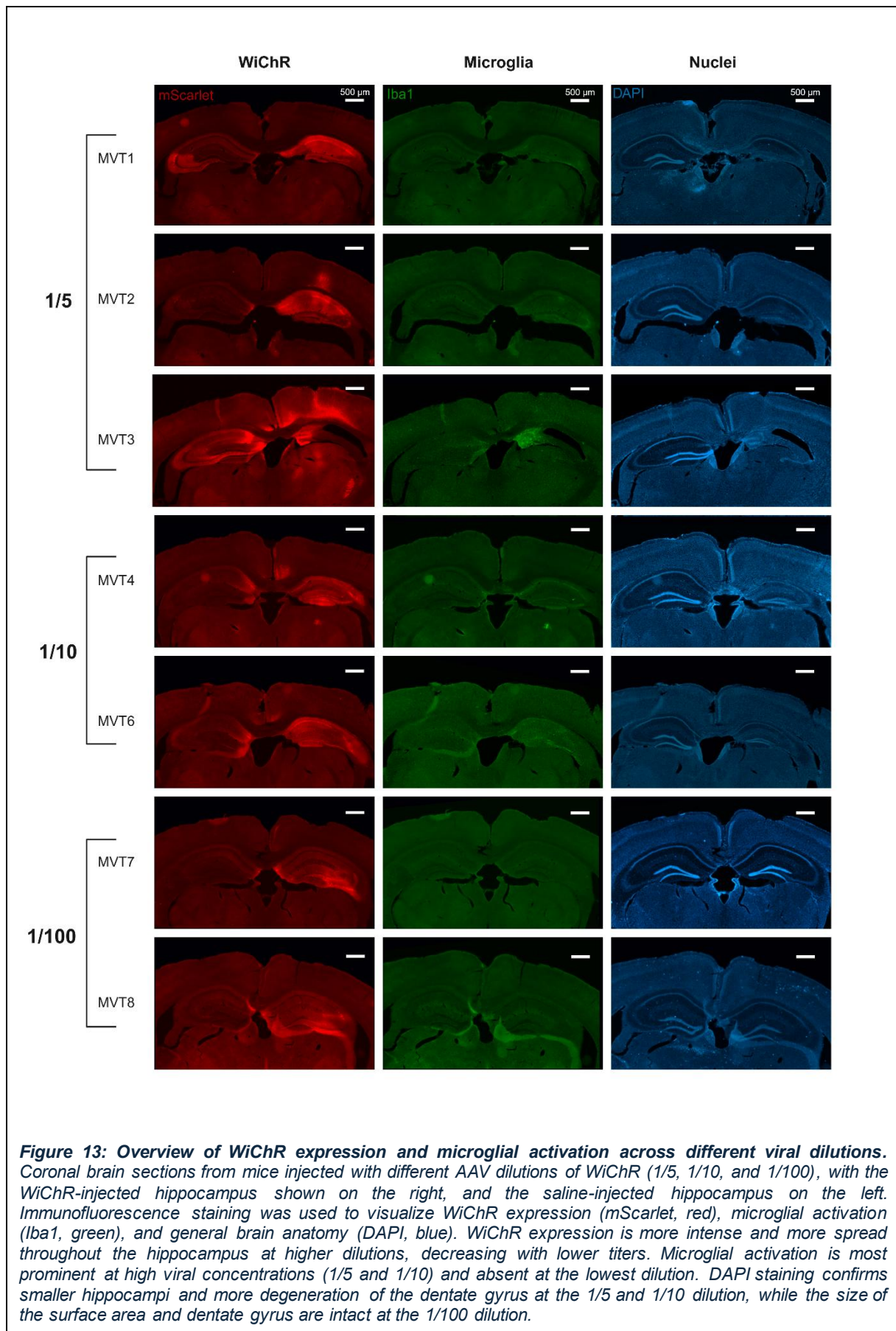
To evaluate the effect of viral titer on neurotoxicity, different dilutions of the AAV2/9.hsyn-WiChR1_ts_mscarlet_er-WPRE titer ($8.47\text{E}+12$ vg/mL) were compared: 1/5, 1/10, and 1/100. The undiluted vector was not included as previous experiments had already shown clear evidence of toxicity. Immunofluorescence staining was used to visualize opsin expression, inflammatory response, and anatomical integrity across conditions. Specifically, brain sections were stained for mScarlet (tagged to WiChR) to assess opsin expression, Iba1 to label microglia and evaluate potential inflammatory responses, and DAPI to stain nuclei to provide a general overview of hippocampal anatomy. Of the initial nine animals, two were excluded from analysis: MVT5 due to oblique splicing and overlap, and MVT9 for use in another experiment (Addendum, Figure II). An overview of all animals for each WiChR dilution is shown in Figure 13.

At the 1/5 dilution (MVT1 – MVT3), Iba1 immunostaining reveals a clear microglial signal, especially around the injection site, indicating an inflammatory response. This microglial activation is most pronounced in MVT3 and is also minimally present in the contralateral hippocampus. DAPI staining demonstrates a visibly reduced hippocampal size across all animals, along with degeneration of the dentate gyrus. The most substantial structural degeneration is observed in MVT3. Concerning the expression patterns, strong mScarlet fluorescence indicating opsin expression is observed throughout the injected hippocampus. In MVT2 and MVT3, expression also extends into surrounding cortical areas. Moreover, both MVT1 and MVT3 show visible expression in the contralateral hippocampus.

In comparison, the Iba1 staining of the 1/10 dilution (MVT4, MVT6) shows a microglial response similar to the 1/5 group. While DAPI staining shows a reduction in hippocampal size, the dentate gyrus structure is better preserved compared to animals receiving the 1/5 dilution. This dilution shows less intense mScarlet fluorescence restricted to the injected hippocampus, still remaining visible throughout the entire structure. Some signal is also visible in adjacent cortical areas near the midline.

At the 1/100 dilution (MVT7, MVT8), Iba1 staining reveals no substantial increase in microglial signal compared to the contralateral hippocampus, and DAPI staining confirms preserved hippocampal surface area, with no visible signs of degeneration. mScarlet fluorescence remains clearly detectable but is more restricted to the targeted hippocampal region. A narrow band of expression is again observed near the cortical midline.

To summarize, the highest titer (1/5 dilution) results in pronounced microglial activation, hippocampal shrinkage, and degeneration of the dentate gyrus. Strong WiChR expression is seen throughout the hippocampus. Similar findings are observed at the 1/10 dilution, although with slightly reduced WiChR signal and better preservation of the dentate gyrus structure. At the lowest titer (1/100 dilution), minimal microglial response and intact hippocampal anatomy are observed, while robust opsin expression is maintained, albeit more confined to the target region. These results suggest a dose-dependent relationship between viral titer, WiChR distribution, inflammation, and structural integrity.



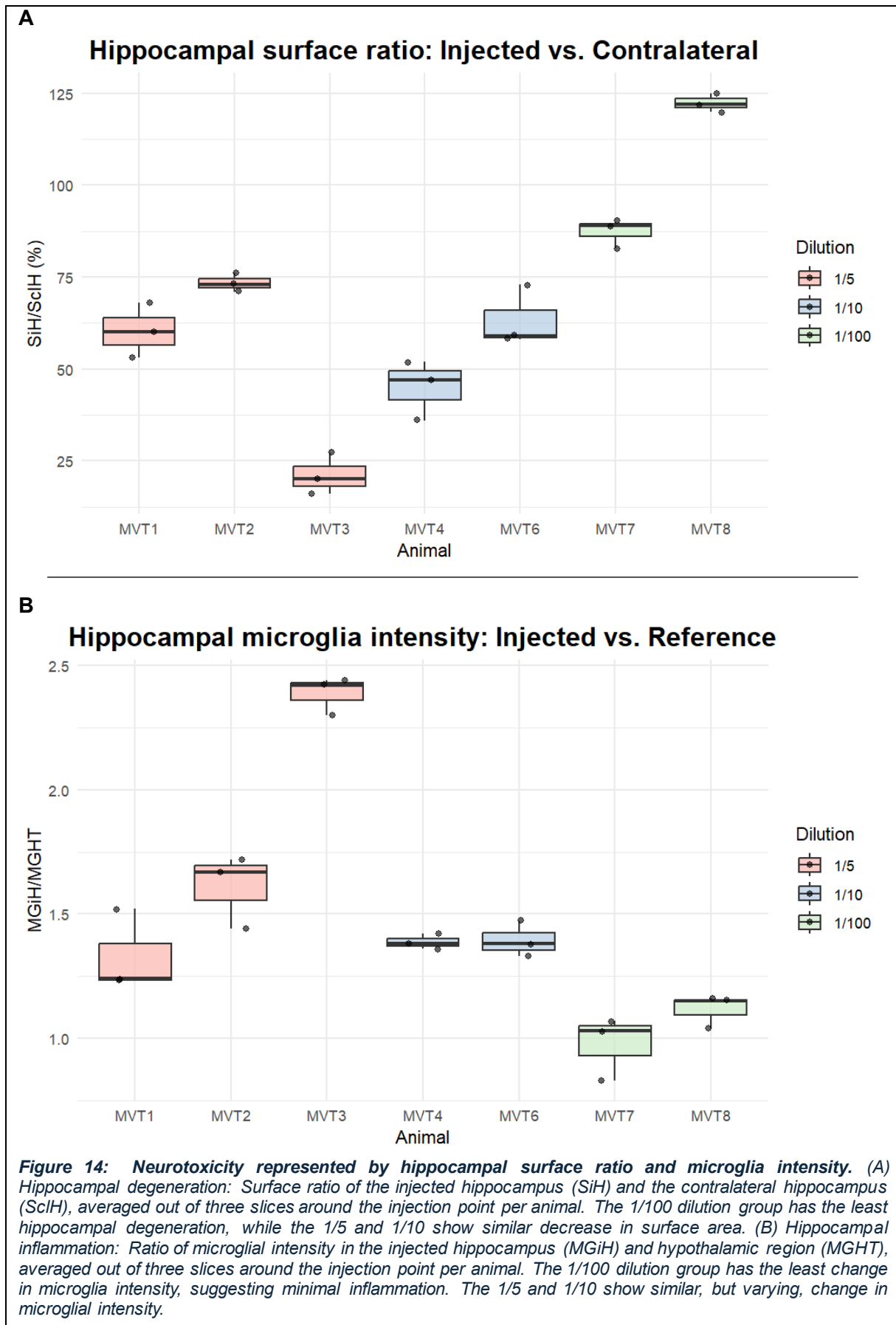
To quantitatively assess neurotoxicity and enable comparison across dilutions, the surface area of the injected hippocampus was measured relative to the contralateral hippocampus, using DAPI-stained images. The inflammatory response was evaluated by calculating the ratio of microglial (Iba1) signal intensity in the injected hippocampus, normalized to a reference region in the hypothalamus to account for inter-slice intensity variability. A comparison of hippocampal surface area ratios and normalized microglial intensities across all animals is shown in Figure 14 (panels A and B, respectively).

In the 1/5 dilution group, MVT1 and MVT2 show a similar decrease in surface area, with MVT1 showing approximately a $40\% \pm 7\%$ decrease in hippocampal surface and MVT2 a $27\% \pm 3\%$ decrease. MVT3 exhibits much more hippocampal degeneration as opposed to the other two animals, with a $79\% \pm 6\%$ decrease in hippocampal surface. On average, the 1/5 dilution leads to a $48\% \pm 27\%$ decrease in hippocampal surface area. When examining microglial intensity, MVT1 and MVT2 show a similar increase in microglial intensity of $33\% \pm 17\%$ and $61\% \pm 15\%$ respectively, compared to the reference region. However, MVT3 has a considerable higher amount of microglia, displaying $139\% \pm 7\%$ more intensity. On average, the 1/5 dilution of the WiChR opsin results in a $78\% \pm 49\%$ higher microglial intensity compared to the reference.

Considering the 1/10 dilution, all animals show more or less similar decrease in surface area, with MVT4 and MVT6 showing respectively a $55\% \pm 8\%$ and $37\% \pm 8\%$ decrease in surface. Altogether, the 1/10 dilution shows a $46\% \pm 13\%$ decrease in hippocampal surface area. Regarding the microglial intensity, both animals show similar results (MVT4: $38\% \pm 3\%$; MVT6: $39\% \pm 7\%$), averaging to $39\% \pm 5\%$ higher intensity compared to the reference region.

Lastly, the 1/100 dilution group shows a $13\% \pm 4\%$ decrease and $22\% \pm 3\%$ increase in hippocampal surface for MVT7 and MVT8, respectively. On average, the 1/100 dilution shows a $5\% \pm 24\%$ increase in hippocampal surface area. As for the microglia, both animals show a low increase in microglial intensity compared to the reference region (MVT7: $-3\% \pm 13\%$; MVT8: $12\% \pm 7\%$), averaging to a $5\% \pm 12\%$ increase.

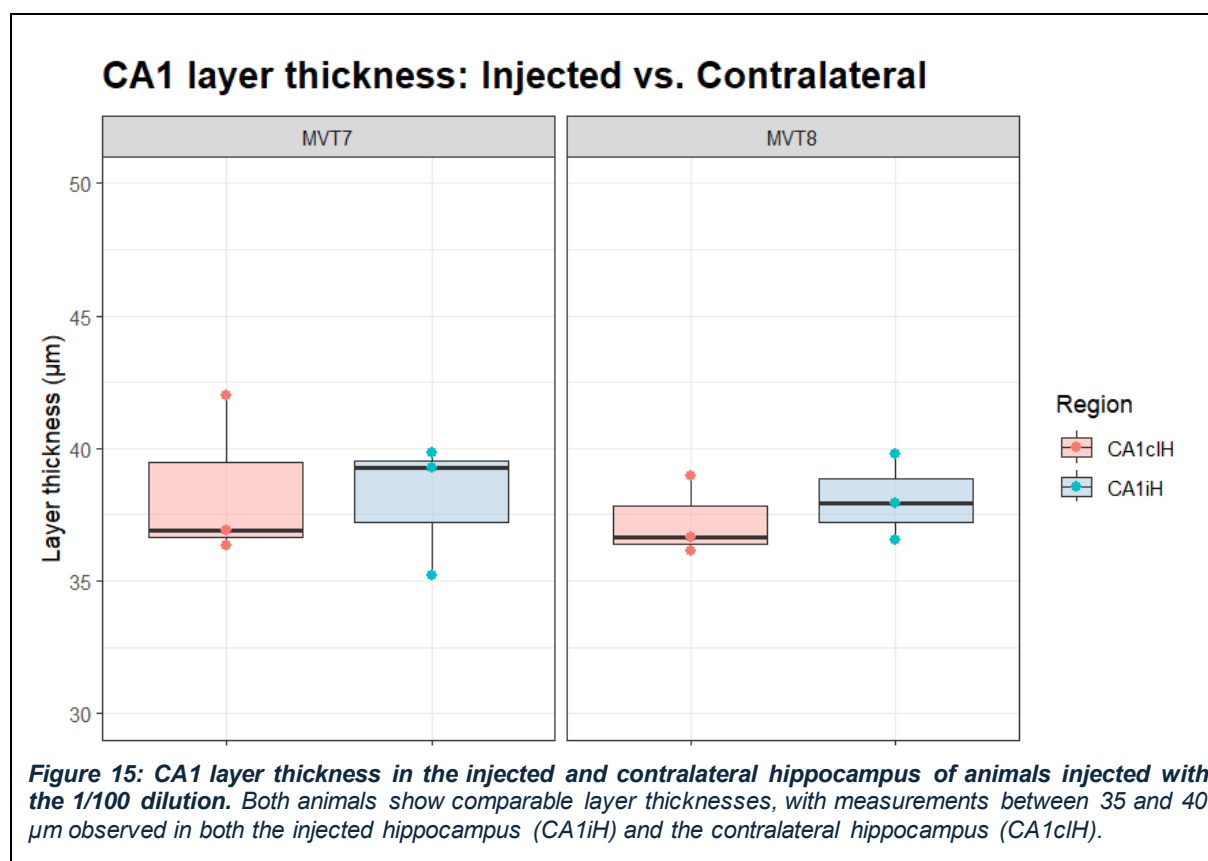
Statistical analysis using the *Kruskal-Wallis* test was conducted to assess whether there is a statistically significant difference between dilutions. This test shows that there is no statistically significant difference in either the hippocampal surface area ($p > 0.05$) and microglial intensity ($p > 0.05$). This can be attributed to the small sample size of seven animals in this small exploratory experiment.



To further confirm the absence of toxicity at the 1/100 dilution, the CA1 pyramidal cell layer thickness was measured on DAPI-stained sections from these animals. The ratio of layer thickness between the injected and contralateral hippocampi was calculated, as shown in Figure 15.

In MVT7, the CA1 layer of the injected hippocampus is $1\% \pm 8\%$ thinner than its contralateral counterpart. In MVT8, the injected hippocampus shows a $2\% \pm 8\%$ increase in CA1 layer thickness. On average, only a 1% difference between both CA1 regions is seen after injection with the 1/100 dilution.

To summarize the results of the dose-titration experiment, both the 1/5 and 1/10 dilution result in notable hippocampal degeneration and increased microglial activation, indicating neurotoxicity and an inflammatory response. While variability is observed – particularly at the 1/5 dilution (Figure 14, MVT3) – the overall trend remains consistent across animals. In contrast, the 1/100 dilution shows minimal change in hippocampal surface area, and no significant increase in microglia, suggesting an absence of toxicity at this concentration. This finding is further supported by the lack of notable differences in CA1 layer thickness between the injected and contralateral hippocampi. Notably, animals with greater hippocampal shrinkage also tend to show higher microglial intensity; for example, MVT3 displays the most severe tissue loss and the strongest inflammatory response. These findings demonstrate a clear dose-dependent relationship, where lower opsin concentrations are associated with reduced degeneration and inflammation. Based on these results, the 1/100 dilution was selected to use in further experiments as no evidence of toxicity could be seen.



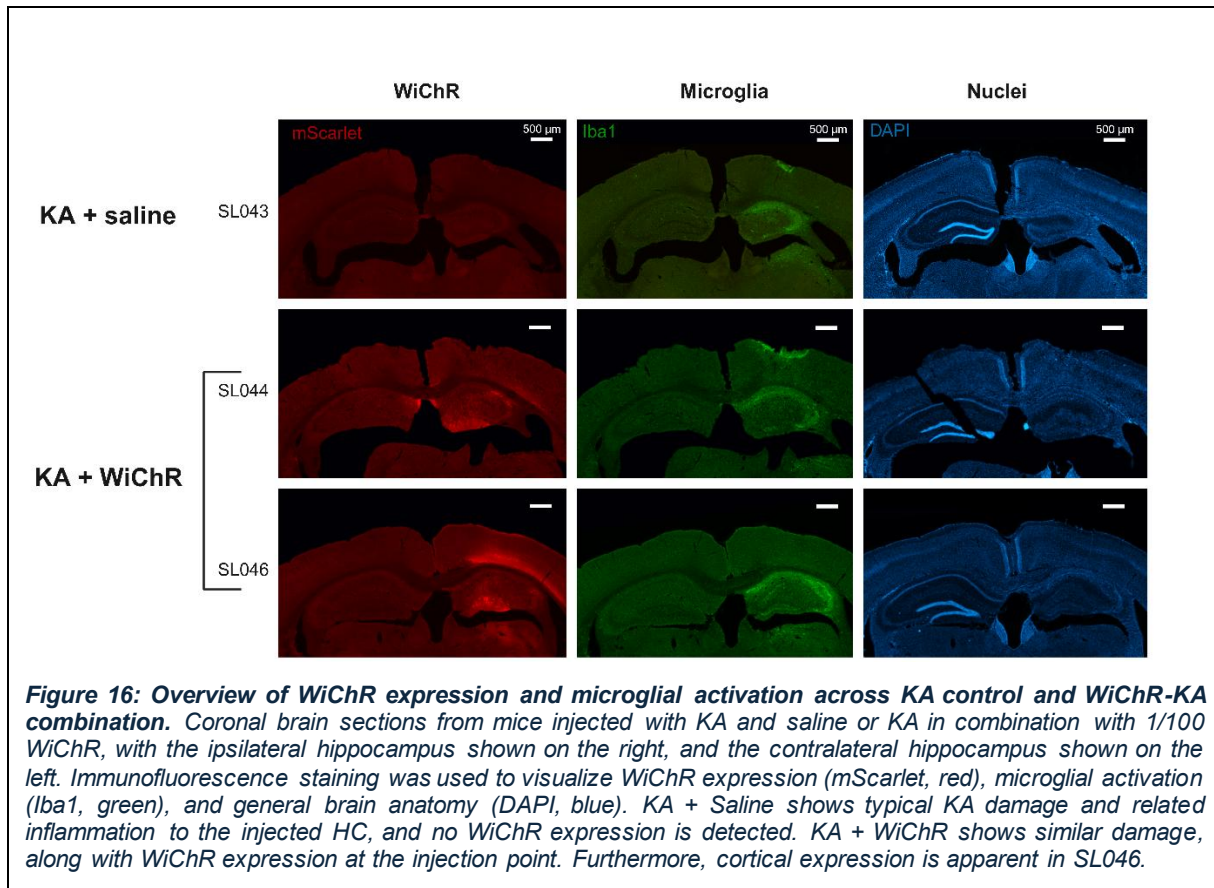
5.1.2 WiChR-KA

The WiChR-KA experiment (Batch 2) was conducted to confirm WiChR expression and non-toxicity of the 1/100 dilution in IHKA animals. Two animals received both WiChR and KA injections. One animal was injected with KA and saline instead of the viral vector to serve as a control. Similar to the previous experiment, immunofluorescence staining was performed to visualize opsin expression, microglial activation, and cell nuclei. Figure 16 shows examples of animals from each experimental condition. Due to the limited sample size, only a qualitative analysis could be performed.

In the KA + saline control animal (SL043), Iba1 staining reveals microglial activation around the pyramidal layer of the KA-injected hippocampus, and in the overlying cortex surrounding the injection site. DAPI staining shows a smaller hippocampus and a loss of typical anatomical organization in the ipsilateral hippocampus, including clear evidence of granule cell dispersion – an abnormal widening of the dentate granule cell layer – as expected in the IHKA mouse model. No WiChR signal is detected in the mScarlet channel as can be expected since no viral vector was injected.

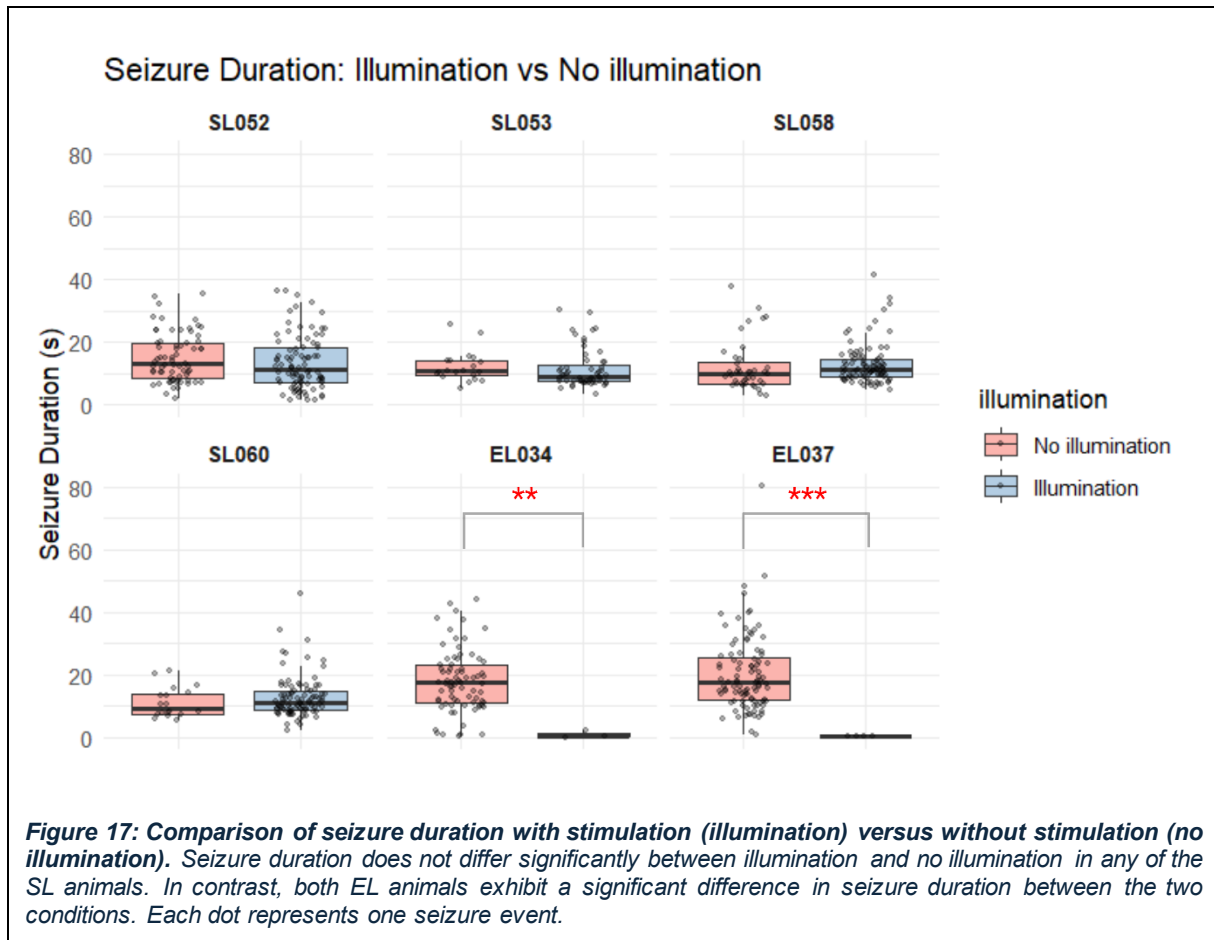
In animals injected with both KA and 1/100 WiChR (SL044 and SL046), Iba1 staining reveals pronounced microglial activation around the pyramidal layer in the injected hippocampus of both animals, with an additional cortical microglial response surrounding the injection site in SL044. DAPI staining indicates hippocampal atrophy and disruption of normal cytoarchitecture in the ipsilateral hippocampus of both animals, again including granule cell dispersion as expected in the IHKA mouse model. mScarlet fluorescence is detected around the injection site and in the overlying cortex, particularly above corpus callosum. This cortical expression is more pronounced in SL046 compared to SL044.

To summarize the histological findings from the WiChR-KA experiment, the KA control animal exhibits an expected inflammatory response and structural degeneration associated with the IHKA model. KA animals injected with 1/100 WiChR show a similar response with localized opsin expression around the injection site. Collectively, these findings suggest that the 1/100 dilution of WiChR provides sufficient expression in KA animals without evidence of additional toxicity attributable to WiChR.



5.2 Responsive Neuromodulation

Following the toxicity study, the efficacy of the responsive neuromodulation system in suppressing seizures was evaluated. To this end, ten animals were injected with KA to induce the occurrence of spontaneous seizures, followed by injection of the WiChR opsin and implantation of the optrode. SL animals (n=10) were injected with viral vector using the Hamilton injector. Three animals died prior to the initiation of recordings, and two other animals did not develop spontaneous seizures, leaving five animals available for analysis. One of these animals, SL055, displayed clear seizures during the recording sessions, but was excluded as the seizures were neither detected nor subjected to optogenetic stimulation due to a technical issue with the seizure detection method (Addendum, Figure III). Figure 17 depicts the seizure durations per animal during recording sessions with and without optogenetic stimulation. It is important to note that the reported seizure durations refer to the duration after detection.



In the absence of optogenetic stimulation, animals SL052, SL053, SL058, and SL060 exhibit mean seizure durations of 15.01 ± 7.77 , 12.23 ± 5.00 , 12.22 ± 8.09 , and 11.13 ± 4.89 seconds, respectively. In contrast, following illumination, the corresponding seizure durations are comparable, with means of 13.35 ± 8.58 , 11.45 ± 6.29 , 12.97 ± 6.44 , and 12.76 ± 6.76 seconds for the same animals. Statistical analysis using the Wilcoxon signed-rank test was performed to determine whether optogenetic stimulation significantly affected seizure duration. No statistically significant differences in seizure duration were observed between the stimulation (illumination) and no stimulation (no illumination) conditions for any of the SL animals (SL052, SL053, SL058, SL060), with all p-values exceeding the 0.05 threshold ($p = 0.12$, 0.13 , 0.06 , and 0.27 , respectively).

Following the absence of an effect on seizure suppression in the SL animals, histological analysis was conducted to investigate the underlying cause. Histological images from these animals are shown in Figure 18. Overall, the histological findings reveal an absence of detectable WiChR expression in the injected hippocampus, which accounts for the lack of seizure suppression. These animals may serve as negative controls to demonstrate that illumination alone has no effect on seizure activity. Additionally, anatomical evaluation of the brain sections from animal SL056 shows the absence of a kainic-acid induced lesion in the injected hippocampus, providing a plausible explanation for the lack of seizure activity in this animal.

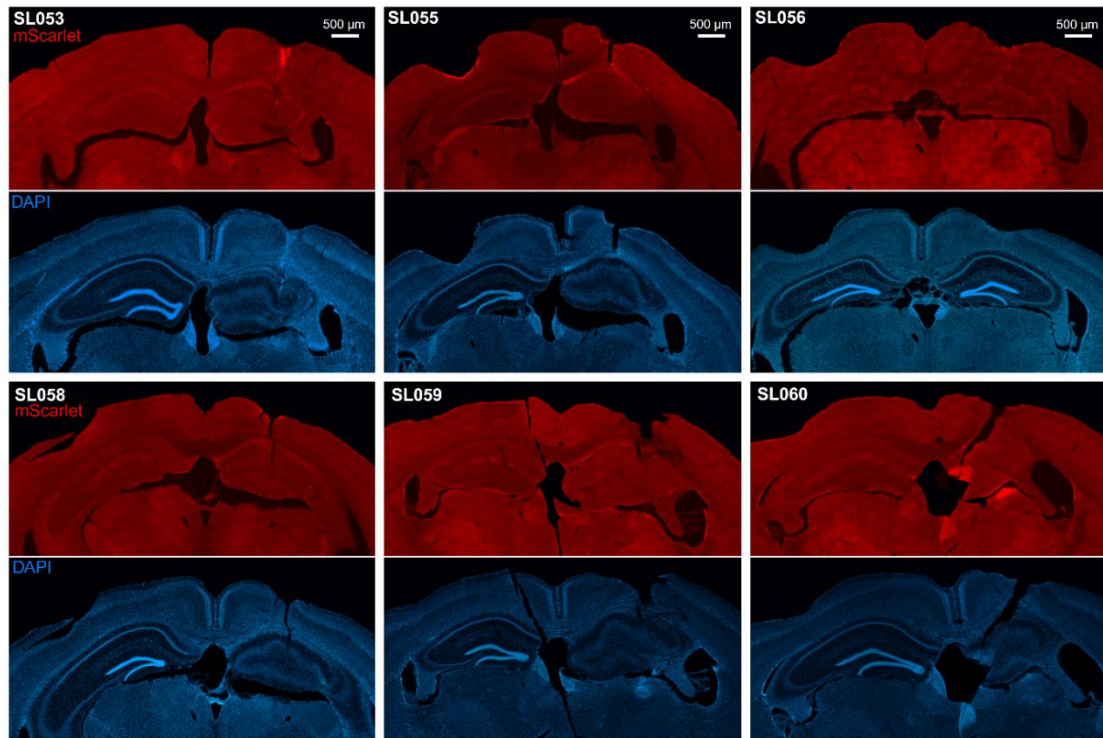


Figure 18: Overview of WiChR expression and anatomy across SL animals. Coronal brain sections from mice injected with kainic acid and 1/100 WiChR, with the ipsilateral hippocampus shown on the right, and the contralateral hippocampus on the left. Immunofluorescence staining was used to visualize WiChR expression (mScarlet, red) and general brain anatomy (DAPI, blue). No notable WiChR expression is observed in the injected hippocampus across all animals. All animals show a typical kainic-acid induced lesion, except for SL056, which has preserved hippocampal anatomy.

A notable procedural change compared to the toxicity study, was the use of a Hamilton syringe instead of a Nanoliter injector for the delivery of WiChR. To test whether this change in injector affected WiChR expression, one healthy animal was injected with WiChR using the Nanoliter injector in the right hippocampus and the Hamilton syringe in the left hippocampus. Histological analysis demonstrated markedly higher WiChR expression following Nanoliter injection compared to the Hamilton injection (Addendum, Figure IV), confirming the hypothesis that the lack of expression in the SL animals from Batch 3 was caused by the change in injection system.

Simultaneously, an additional experiment was set up with two IHKA animals from another experiment (EL034, EL037). These animals received WiChR injections using the Nanoliter injector instead of the Hamilton injector, followed by implantation and EEG recordings. The results are also shown in Figure 17. EL034 exhibits a mean seizure duration of 17.90 ± 9.93 seconds without optogenetic stimulation, which was markedly reduced to 0.91 ± 1.29 seconds after illumination. Similarly, EL037 shows a reduction from 20.15 ± 9.93 seconds without illumination to 0.26 ± 0.16 seconds with stimulation. Respectively, a reduction of $95\% \pm 1\%$ and $99\% \pm 1E-05$ in seizure duration is observed. Statistical analysis using the Wilcoxon signed-rank test confirms a significant reduction in seizure duration between the stimulated and non-stimulated conditions (EL034: $p = 0.006^{**}$; EL037: $p = 7.51E-04^{***}$), demonstrating successful seizure suppression following detection and during illumination.

To summarize, the SL animals show no significant reduction in seizure duration after optogenetic stimulation, explained by the absence of WiChR expression. In contrast, two EL animals injected with WiChR via the Nanoliter injector show significant seizure duration reductions after stimulation, demonstrating effective seizure suppression as the seizures were arrested almost instantaneously.

6 Discussion

This thesis aimed to evaluate the safety and efficacy of the WiChR opsin in a preclinical mouse model for temporal lobe epilepsy. The investigation first focused on identifying a non-toxic viral dilution of WiChR in healthy mice (dose-titration), followed by an evaluation in an IHKA mouse model (WiChR-KA). Subsequently, the efficacy of this non-toxic dilution in suppressing seizures was examined in KA mice using a responsive neuromodulation system.

6.1 Neurotoxicity

6.1.1 Dose-titration

The dose-titration experiment assessed the neurotoxic effects of varying AAV-mediated WiChR opsin concentrations in the mouse hippocampus by evaluating hippocampal degeneration and microglial activation. The results revealed a dose-dependent relationship between viral titer, tissue degeneration, and inflammatory response. The two highest titers (1/5 and 1/10) produced comparable degrees of hippocampal shrinkage and microglial activation, whereas the lowest titer (1/100) did not result in observable hippocampal surface reduction or evidence of an inflammatory response.

The small difference between the 1/5 and 1/10 dilutions likely reflects the modest two-fold change in concentration, compared to the larger 10- and 20- fold differences when comparing these to the 1/100 dilution. Increasing the number of dilution groups or distributing the concentrations more evenly – such as 1/10, 1/50, and 1/100 – could improve dosing precision. For instance, Stevens et al. employed seven CAV2-DREADD groups with varying dilutions and injection volumes to assess both toxicity and expression⁴⁸. While such designs increase insight, they also demand larger sample sizes. This study aimed to provide an estimate of safety and efficacy, while balancing ethical, scientific, and practical considerations.

Despite the large differences between the lowest dilutions and the highest dilution, none were statistically significant due to the small sample size and large inter-animal variability. Animal MVT3, in particular, showed a markedly greater reduction in hippocampal surface area and increase in microglial intensity compared to other animals in this group. This is likely due to biological variability in response to the vector or procedure. Even among inbred strains, immune responses to AAV injections can vary due to subtle epigenetic differences, genetic polymorphisms, variability in gut microbiota, and environmental factors^{49, 50}. No procedural deviations or health abnormalities were noted for MVT3, making technical causes unlikely.

Furthermore, across all groups, hippocampal degeneration showed greater variability than microglial intensity, likely due to the susceptibility of surface area measurements to both technical and biological factors, including sectioning angle, individual anatomical variation, and tissue processing artifacts such as shrinkage or distortion⁵¹. In contrast, microglial activation was quantified as the percentage of fluorescence intensity within a defined region of interest. This renders it less sensitive to differences in surface area caused by uneven or oblique slicing.

These results support the overall hypothesis that high titers of WiChR vector can trigger neuroinflammation and structural degeneration, either through high transgene mRNA/protein expression, or immune responses caused by the AAV vector itself⁵². First, in the case of WiChR, beyond the potential effects of protein overexpression, the microbial origin of the opsin increases the likelihood of immune recognition in mammalian tissue. In addition, WiChR is fused to a fluorescent reporter to enable visualization, thereby increasing the overall size of the protein and potentially enhancing its immunogenicity and cellular burden. If WiChR were to be translated into a therapeutic application, removal of this fluorescent tag would be necessary to reduce these risks. Second, although AAV vectors are widely employed in preclinical neuroscience research and are used in clinically approved gene therapies such as Luxturna®, their safety within the CNS remains incompletely characterized, and instances of neurotoxicity have been reported. Notably, neuronal progenitor cell death has been observed in mice following AAV administration at concentrations ranging from 1.5×10^{12} and 3.6×10^{13}

vg/mL – a range that encompasses the original titer of the WiChR vector used in this study ($8.47\text{E}+12$ vg/mL)⁵³. This raises the possibility that the observed toxic effects may be attributed primarily to the high viral load rather than the transgene expression itself.

A notable finding in this study is the apparent relationship between hippocampal shrinkage and microglial activation. Animals with more pronounced tissue loss, also exhibited stronger Iba1 staining, indicating heightened inflammatory response. The relationship between neurodegeneration and microglia activation has been well established in neurodegenerative diseases such as Parkinson's disease and Alzheimer's disease⁵⁴. While the complex pathogenesis of these conditions lies beyond the scope of this study, they highlight the critical role of microglia in mediating neuronal damage. By inducing neuroinflammation and using a time-resolved histological approach, Jung et al. showed that microglial activation preceded neuronal aberrations, suggesting a causative role of microglia in neurotoxicity⁵⁵. This indicates that toxic agents may exert their detrimental effects indirectly, by triggering microglial responses rather than acting directly on neurons. Assessing additional markers, such as vimentin for astrocytes (e.g. vimentin) or TNF- α for peripheral immune involvement, could help elucidate the involvement of multiple cellular pathways⁴².

Although opsin expression was not extensively analyzed, some characteristics of the expression could be observed. At higher titers (1/5 and 1/10), strong WiChR expression was observed, with widespread distribution throughout the hippocampus and occasional extension into cortical areas. Cortical expression – most prominent at the highest titer and diminishing with lower concentrations – may result from unintended leakage of the viral vector along the needle tract or the natural cortical tropism of AAV9⁵⁶. Moreover, contralateral hippocampal expression – particularly at high titers – suggests possible axonal or transsynaptic transport via commissural fibers connecting both hippocampi^{35, 57}. At the lower titer (1/100), expression was more confined to the targeted hippocampal region, with cortical expression and contralateral expression nearly absent compared to higher titers. This indicates reduced viral spread and a more localized transduction pattern, minimizing off-target expression.

Study limitations

Even though this toxicity study identified a safe opsin dose that does not induce significant toxicity in healthy mice, several limitations should be acknowledged. First, this was a small-scale, exploratory study. While qualitative findings are robust and indicate a dose-dependent trend, the limited number of animals restricted statistical generalizability and increased the susceptibility to individual variability. Based on effect sizes from two previous Ghent University studies by Goossens et al. and Stevens et al., and calculations using G*Power (v3.1.9.7), a sample size of 30 animals (10 per dilution) would be required to achieve 80% power at a significance level (α) of 0.05 for comparing dilutions reliably^{39, 48}.

Additionally, the analytical methods in this study were limited to an analysis of histopathological changes in brain slices, which only provide limited information on the status of the brain and are vulnerable to oblique sectioning. The study could be improved by including MRI-based volumetric imaging to provide more accurate and comprehensive measurements of hippocampal size³⁹. Additionally, behavioral assessments should be conducted to investigate potential cognitive or functional toxicity⁵⁸. Such assessments are essential for determining functional safety, as emphasized by the International Conference on Harmonization (ICH), which states that the effects of a test substance on the CNS should be thoroughly assessed⁵⁹. These evaluations should include motor activity, behavioral changes, coordination, sensory and motor reflexes, and body temperature. Another unresolved issue is whether observed toxicity arises primarily from the viral load or from WiChR expression itself. Clarifying this distinction is critical for clinical translation and could be addressed by employing control AAV vectors lacking a transgene, as demonstrated by Goossens et al., or by using alternative non-viral delivery methods³⁹. Finally, this study did not include a quantitative assessment of viral spread, limiting insights into the precise distribution and extend of WiChR expression across brain regions, which is vital information to assess the efficacy of the WiChR opsin in future studies.

6.1.2 WiChR-KA

Following the dose-titration, the WiChR-KA experiment assessed whether the 1/100 WiChR dilution remains non-toxic and provides expression in the IHKA mouse model, given that KA-induced pathology may increase tissue susceptibility compared to healthy animals. The results demonstrated that the 1/100 dilution achieved visible hippocampal opsin expression in KA-injected animals without exacerbating neuroinflammatory responses or structural degeneration beyond those induced by KA alone. This supports the safety and suitability of this dilution for further experiments.

Study limitations

The limitations of this experiment largely mirror those of the dose-titration study, including the qualitative nature of the toxicity assessment and the use of only three animals. No quantitative analysis was performed for this experiment, as it was intended to provide a quick initial impression of toxicity in the IHKA mouse model. To draw firm conclusions on whether WiChR expression adds toxicity beyond that of KA alone, a statistically powered study would be required of approximately 60 animals. However, the primary objective here was to obtain an initial impression, which did not necessitate a larger cohort and thereby reducing the number of animals used.

6.2 Responsive Neuromodulation

The second part of this thesis aimed to evaluate the efficacy of responsive neuromodulation (RNM) using optogenetics to suppress seizures in a mouse model of TLE. Initial trials using a Hamilton syringe for opsin delivery failed to produce significant seizure suppression, which histological analysis later attributed to insufficient WiChR expression. In contrast, follow-up experiments using a Nanoliter injector demonstrated stronger opsin expression compared to the Hamilton injection, highlighting the importance of vector delivery technique in achieving therapeutic outcomes. Following correction of the injection method, seizures could be suppressed almost instantaneously, leading to a dramatic reduction in seizure duration compared to seizures without intervention. This provides strong evidence for the therapeutic potential of RNM.

Despite these promising results, the conclusions should be interpreted with caution, as none of the initial 10 animals exhibited seizure suppression following illumination. This was hypothesized to be caused by a problem with the injection system. This change from the toxicity study was made due to an unusually low success rate of the KA injections with the Nanoliter injection in parallel experiments. A possible cause could be an increase in KA toxicity due to the small diameter of the tip of the pipet in the Nanoliter injector. The Hamilton injector has a wider tip and was therefore hypothesized to be better suited for small volumes like use in the KA injections; in principle, this should not have made a difference for larger volumes, such as the vector. Therefore, a small experiment was performed on two animals injected with the Nanoliter. These animals did show clear and instantaneous seizure suppression, confirming the efficacy of the WiChR opsin despite the diluted concentration. However, a sample size of two animals is not sufficient to draw statistically supported conclusions, limiting the generalizability of the findings.

The use of optogenetics in RNM offers several key advantages, including cell-type specificity and precise spatial and temporal control. Unlike electrical stimulation, it enables targeted modulation of neuronal populations without nonspecific activation of surrounding brain regions, thereby minimizing off-target effects and network destabilization. However, the main advantage of optogenetics – its cell-type specificity – remains underutilized in this study. The use of the broad hSYN promoter, while neuron-specific, results in non-selective suppression of both excitatory and inhibitory neurons. This generalized suppression may not optimize therapeutic outcomes, and future approaches should explore more targeted strategies – particularly selective suppression of excitatory neurons, as demonstrated by Krook-Magnuson et al. – to enhance efficacy⁶⁰. Nevertheless, as highly specific promoters often exceed AAV packaging limits, alternative vectors – such as lentiviruses or adenoviruses – should be

considered. The use of these vectors would necessitate renewed toxicological evaluation, as they differ in their biological properties. Additionally, optimal dosing should be explored through high-sample, dose-titration studies – beginning with conservative doses to assess safety and efficacy, followed by dose-escalation in approximately 10-fold increments – based on the conclusions of the current study. Nevertheless, despite this lack of specificity, effective seizure suppression was still achieved. This suggests that precise neuronal targeting may not be a strictly necessary for functional efficacy, potentially offering a more flexible and robust therapeutic strategy in translational applications where cell-type specificity is harder to achieve.

Future perspectives

Future studies will aim to validate seizure suppression using the WiChR opsin by expanding sample size. Additionally, parameter exploration will be performed to find the most efficient stimulation protocol. The stimulation parameters in the current study were selected to ensure observable effects. Expanding beyond the seizure suppression analysis, future pre-clinical studies should include behavioral studies to verify potential side effects of the stimulation, as well as the potential benefits from seizure suppression. Prior to initiating clinical trials, additional validation in non-human primates, human organotypic tissue, and in silico modeling is recommended to better reflect human application of the WiChR opsin and the implantable device. Only after thorough pre-clinical validation of safety and efficacy, translation to clinical practice can be explored. Importantly, this approach involves the use of a Gene Therapy Medicinal product (GTMP), which raises ethical concerns that must be thoroughly addressed⁶¹. A pioneering therapy in this field is the use of optogenetics in human patients with Retinitis Pigmentosa⁶². This therapy paves the way to acceptance in other vector-based therapies. Additionally, the system employs an Active Implantable Medical Device (AIMD), introducing further regulatory complexity and associated ethical considerations. Nevertheless, the promising therapeutic potential of this approach for treating refractory temporal lobe epilepsy (TLE) highlights its relevance for future clinical development.

7 General Conclusion

This study systematically evaluated the safety and efficacy of the WiChR opsin in a mouse model of temporal lobe epilepsy. Firstly, three dilutions of the WiChR vector (1/5, 1/10, or 1/100) were tested for neurotoxicity. The results revealed clear dose-dependent neurotoxic effects, with higher WiChR concentrations (1/5, 1/10) resulting in significant hippocampal degeneration and microgliosis, while no toxic effects were seen in lower WiChR vector concentrations (1/100). Subsequent administration of this non-toxic WiChR dilution in the kainic acid mouse model revealed no additional toxic effects, indicating the 1/100 dilution is safe for use in this model.

Secondly, the 1/100 dilution of the WiChR vector was employed to assess its efficacy in suppressing seizures using a responsive neuromodulation system. Initial experiments using Hamilton injector failed to elicit an effect, attributed to insufficient WiChR expression due to a technical problem with the injections. Upon repeating the experiment with the Nanoliter injector, a significant reduction in seizure duration was observed after illumination.

In summary, this study identified a safe and effective WiChR vector concentration that enables optogenetic suppression in a preclinical mouse model of temporal lobe epilepsy, thereby contributing a promising step toward the development of targeted neuromodulation therapies.

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9 Statement on the use of Generative AI-tools

During this thesis, AI chat tools (ChatGPT, DeepSeek, Anara) were used for writing support – such as clarifying statistical concepts and improving phrasing – but not for generating scientific content, data, or interpretations; all analyses and conclusions remain our own and were carefully verified.



Investigating the toxicity of WiChR vector in responsive neuromodulation as therapy for epilepsy

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Introduction

- 30% of epilepsy patients are **non-responsive to anti-seizure drugs**.
- **Responsive neuromodulation** is a novel approach where seizures are suppressed in real-time.
- In this study, **virally introduced light-sensitive opsins** are activated with an implanted light emitting probe to suppress seizures (optogenetics).
- The state-of-the-art **WiChR opsin** unlocks new possibilities, due to its **K⁺-selectivity**, provoking repolarization during a seizure.

WiChR's safety must be validated before full implementation in the kainic acid (KA) mouse model for temporal lobe epilepsy (TLE).

We assessed **hippocampal toxicity** after injections with different dilutions (1/5, 1/10, 1/100) without KA by measuring hippocampal surface area and quantifying microglia. The least toxic dilution will be further explored for optogenetic epilepsy treatment.

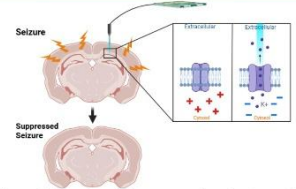


Figure 1: Mechanism of seizure suppression via optogenetics. 470nm LED opens K⁺-selective WiChR, provoking repolarization and suppressing seizure.

Methods

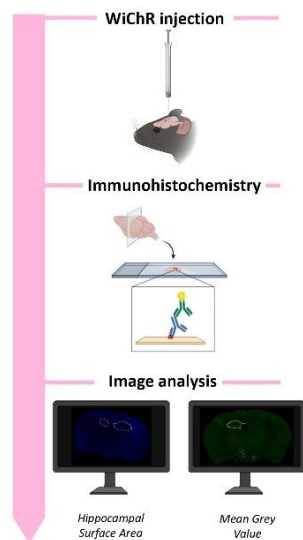
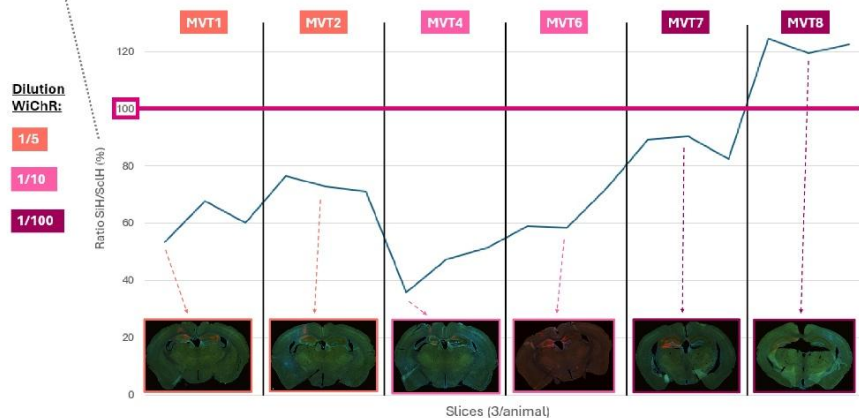


Figure 2: Timeline of experiment

Results

$$\frac{SH}{ScH} = \frac{\text{Surface injected Hippocampus}}{\text{Surface contralateral Hippocampus}}$$

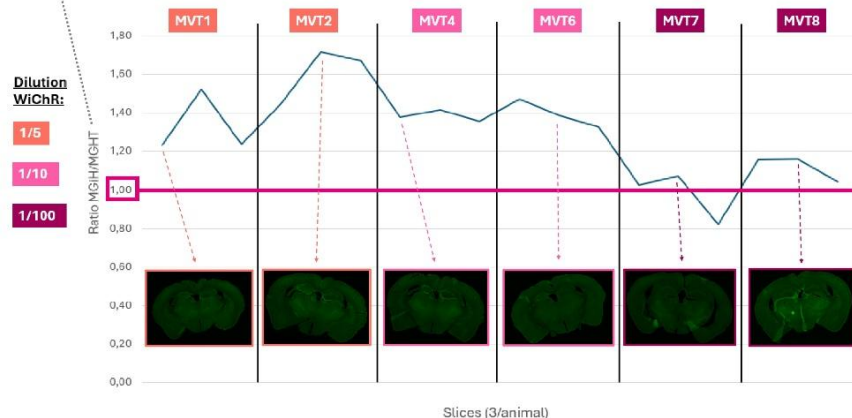
Surface ratio: Hippocampal degeneration



A low ratio of injected to contralateral hippocampus indicates a smaller surface of the injected hippocampus, representing a high hippocampal degeneration. The graph indicates that lower WiChR concentrations show a decreased difference in surface area, and thus lower toxicity. **The 1/100 dilution shows little to no degeneration of the injected hippocampus.**

$$\frac{MGI}{MCHT} = \frac{\text{Microglia injected Hippocampus}}{\text{Microglia Hypothalamic region (reference)}}$$

Microglia: Hippocampal inflammation



The normalized amount of microglia represents inflammation in the injected hippocampus. Inflammation increases after opsin injection: Lower WiChR concentrations show a decrease in inflammation. **The 1/100 dilution shows the least inflammation in the injected hippocampus.**

Conclusion

Both **hippocampal surface area** and **microglia quantity** appear to correlate with the **dilution of the WiChR** opsin. The more WiChR is diluted, the less degeneration of the injected hippocampus, and the fewer microglia. However, there is insufficient data to state a statistically significant difference in toxicity between the dilutions.

The 1/100 dilution shows little to no toxicity and will be used in further experiments

Future direction

Testing the efficacy of the responsive neuromodulation system to suppress seizures in a KA mouse model.

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- Figures created with BioRender.com

11 Addendum

Potential Misinterpretation of Seizure Termination Due to Coincidental Onset with Stimulation

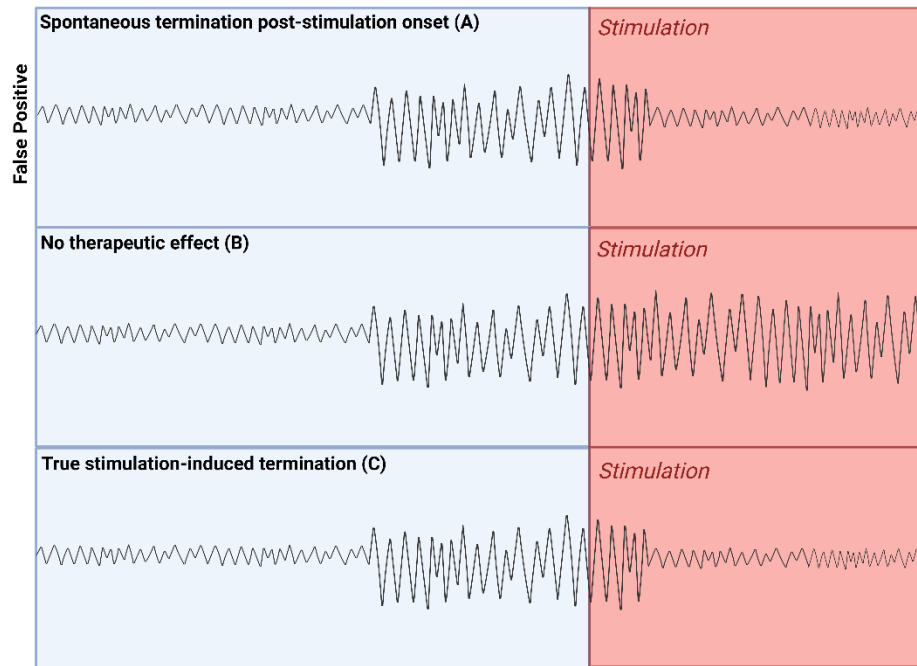


Figure I: Seizures that naturally resolve near stimulation onset may falsely appear suppressed by stimulation. Panel A: The seizure is naturally terminated during stimulation, creating the misinterpretation that the seizure was suppressed by stimulation. Panel B: The stimulation has no therapeutic effect. The seizure is not suppressed during stimulation. Panel C: The seizure is suppressed during stimulation.

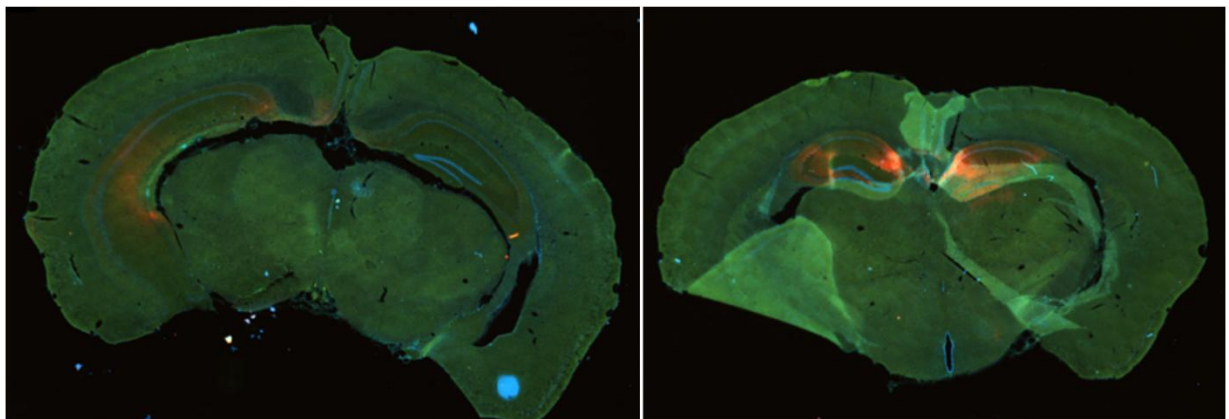


Figure II: Example slice of MVT5. The left hippocampus appears markedly larger than the right due to oblique slicing (left panel). Efforts to align the left hippocampus with the anatomically correct right hippocampus resulted in excessive overlap at the presumed injection point. MVT5 was therefore excluded from analysis.

